

Question 1 Correct

A company would like to define a set of rules to manage objects cost-effectively between Amazon Simple Storage Service (Amazon S3) storage classes. As a Cloud Practitioner, which Amazon S3 feature would you use?

Amazon S3 Transfer Acceleration (Amazon S3TA)

Your answer is correct

Amazon Simple Storage Service (Amazon S3) Lifecycle configuration

S3 Cross-Region Replication (S3 CRR)

Amazon Simple Storage Service (Amazon S3) Bucket policies

Overall explanation

Correct option:

Amazon Simple Storage Service (Amazon S3) Lifecycle configuration

To manage your objects so that they are stored cost-effectively throughout their lifecycle, configure their Amazon S3 Lifecycle. An S3 Lifecycle configuration is a set of rules that define actions that Amazon S3 applies to a group of objects. There are two types of actions: Transition actions (define when objects transition to another storage class) and expiration actions (define when objects expire. Amazon S3 deletes expired objects on your behalf).

In this particular use case, you would use a transition action.

Incorrect options:

Amazon S3 Transfer Acceleration (Amazon S3TA) - Amazon S3 Transfer

Acceleration (Amazon S3TA) enables fast, easy, and secure transfers of files over long distances between your client and an Amazon S3 bucket. It is not used to move objects between storage classes.

Amazon Simple Storage Service (Amazon S3) Bucket policies - An S3 bucket policy is a resource-based AWS Identity and Access Management (IAM) policy. You add a bucket policy to a bucket to grant other AWS accounts or IAM users access permissions for the bucket and the objects in it. Object permissions apply only to the objects that the bucket owner creates. It is not used to move objects between storage classes.

S3 Cross-Region Replication (S3 CRR) - S3 Cross-Region Replication (S3 CRR) enables automatic, asynchronous copying of objects across Amazon S3 buckets.

Cross-Region replication (CRR) is used to copy objects across Amazon S3 buckets in different AWS Regions. It is not used to move objects between storage classes.

References:

<https://docs.aws.amazon.com/AmazonS3/latest/dev/object-lifecycle-mgmt.html>

<https://aws.amazon.com/s3/>

Domain

Technology

Question 2Correct

Which AWS tool can provide best practice recommendations for performance, service limits, and cost optimization?

Amazon CloudWatch

AWS Health Dashboard - Service health

Amazon Inspector

Your answer is correct

AWS Trusted Advisor

Overall explanation

Correct option:

AWS Trusted Advisor

AWS Trusted Advisor is an online resource to help you reduce cost, increase performance, and improve security by optimizing your AWS environment. AWS Trusted Advisor provides real-time guidance to help you provision your resources following AWS best practices.

How AWS Trusted Advisor works:



via - <https://aws.amazon.com/premiumsupport/technology/trusted-advisor/>

Incorrect options:

Amazon Inspector - Amazon Inspector is an automated security assessment service that helps improve the security and compliance of applications deployed on your Amazon EC2 instances. Amazon Inspector automatically assesses applications for exposure, vulnerabilities, and deviations from best practices. Nevertheless, it does not provide best practice recommendations.

AWS Health Dashboard - Service health - AWS Health Dashboard - Service health publishes most up-to-the-minute information on the status and availability of all AWS services in tabular form for all Regions that AWS is present in. It does not provide best practice recommendations.

Amazon CloudWatch - Amazon CloudWatch is a monitoring and observability service built for DevOps engineers, developers, site reliability engineers (SREs), and IT managers. Amazon CloudWatch provides data and actionable insights to monitor applications, respond to system-wide performance changes, optimize resource utilization, and get a unified view of operational health. This is an excellent service for building Resilient systems. Think resource performance monitoring, events, and alerts; think Amazon CloudWatch. Amazon CloudWatch does not provide best practice recommendations.

Reference:

<https://aws.amazon.com/premiumsupport/technology/trusted-advisor/>

Domain

Security and Compliance

Question 3Correct

A company would like to reserve Amazon Elastic Compute Cloud (Amazon EC2) compute capacity for three years to reduce costs. The company also plans to increase their workloads during this period. As a Cloud Practitioner, which Amazon Elastic Compute Cloud (Amazon EC2) reserved instance (RI) type would you recommend?

Your answer is correct

Convertible reserved instance (RI)

Scheduled reserved instance (RI)

Standard reserved instance (RI)

Adaptable reserved instances (RI)

Overall explanation

Correct option:

Convertible reserved instance (RI)

Purchase convertible reserved instance (RI) if you need additional flexibility, such as the ability to use different instance families, operating systems, or tenancies over the reserved instance (RI) term. Convertible reserved instance (RI) provides you with a significant discount (up to 54%) compared to an on-demand instance and can be purchased for a 1-year or 3-year term.

Convertible reserved instance (RI) can be useful when workloads are likely to change. In this case, a convertible reserved instance (RI) enables you to adapt as needs evolve while still obtaining discounts and capacity reservation.

Amazon EC2 Pricing Options Overview:

On-Demand

With On-Demand instances, you pay for compute capacity by the hour or the second depending on which instances you run. No longer-term commitments or upfront payments are needed. You can increase or decrease your compute capacity depending on the demands of your application and only pay the specified per hourly rates for the instance you use.

On-Demand instances are recommended for:

- Users that prefer the low cost and flexibility of Amazon EC2 without any up-front payment or long-term commitment
- Applications with short-term, spiky, or unpredictable workloads that cannot be interrupted
- Applications being developed or tested on Amazon EC2 for the first time

[See On-Demand pricing »](#)

Spot instances

Amazon EC2 Spot instances allow you to request spare Amazon EC2 computing capacity for up to 90% off the On-Demand price. [Learn More.](#)

Spot instances are recommended for:

- Applications that have flexible start and end times
- Applications that are only feasible at very low compute prices
- Users with urgent computing needs for large amounts of additional capacity

[See Spot pricing »](#)

Savings Plans

Savings Plans are a flexible pricing model that offer low prices on EC2 and Fargate usage, in exchange for a commitment to a consistent amount of usage (measured in \$/hour) for a 1 or 3 year term.

Dedicated Hosts

A Dedicated Host is a physical EC2 server dedicated for your use. Dedicated Hosts can help you reduce costs by allowing you to use your existing server-bound software licenses, including Windows Server, SQL Server, and SUSE Linux Enterprise Server (subject to your license terms), and can also help you meet compliance requirements. [Learn more.](#)

- Can be purchased On-Demand (hourly).
- Can be purchased as a Reservation for up to 70% off the On-Demand price.

[See Dedicated pricing »](#)

Reserved Instances

Reserved Instances provide you with a significant discount (up to 75%) compared to On-Demand instance pricing. In addition, when Reserved Instances are assigned to a specific Availability Zone, they provide a capacity reservation, giving you additional confidence in your ability to launch instances when you need them.

For applications that have steady state or predictable usage, Reserved Instances can provide significant savings compared to using On-Demand instances. See [How to Purchase Reserved Instances](#) for more information.

Reserved Instances are recommended for:

- Applications with steady state usage
- Applications that may require reserved capacity
- Customers that can commit to using EC2 over a 1 or 3 year term to reduce their total computing costs

via - <https://aws.amazon.com/ec2/pricing/>

Incorrect options:

Standard reserved instance (RI) - Standard reserved instance (RI) provides you with a significant discount (up to 72%) compared to on-demand instance pricing, and can be purchased for a 1-year or 3-year term. Standard reserved instance (RI) do not offer as much flexibility as convertible reserved instance (RI), such as not being able to change the instance family type; and therefore are not best-suited for this use case.

Review the differences between standard reserved instance (RI) and convertible reserved instance (RI):

<https://docs.aws.amazon.com/whitepapers/latest/cost-optimization-reservation-models/standard-vs.-convertible-offering-classes.html>

Scheduled reserved instance (RI) - AWS does not support scheduled reserved instance (RI), so this option is ruled out.

Adaptable reserved instances (RI) - Adaptable reserved instance (RI) is not a valid type of reserved instance (RI). It is a distractor.

Reference:

<https://aws.amazon.com/ec2/pricing/reserved-instances/pricing/>

Domain

Billing and Pricing

Question 4Correct

According to the AWS Shared Responsibility Model, which of the following are the responsibilities of AWS? (Select two)

Configuring IAM Roles

Encrypting application data

Installing security patches of the guest operating system (OS)

Your selection is correct

Data center security

Your selection is correct

Network operability

Overall explanation

Correct options:

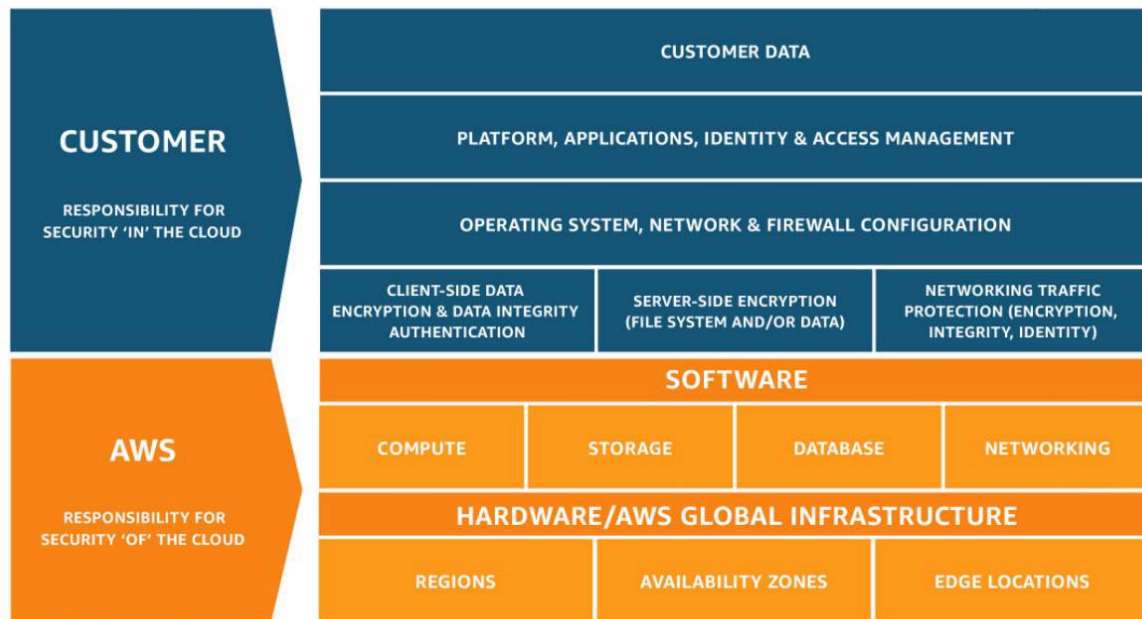
Data center security

Network operability

AWS responsibility "Security OF the Cloud" - AWS is responsible for protecting the infrastructure that runs all of the services offered in the AWS Cloud. This infrastructure is composed of the hardware, software, networking, and facilities that run AWS Cloud services.

Exam Alert:

Please review the AWS Shared Responsibility Model in detail as you can expect multiple questions on the shared responsibility model in the exam:



via - <https://aws.amazon.com/compliance/shared-responsibility-model/>

Incorrect options:

Installing security patches of the guest operating system (OS) - The customers are responsible for patching their guest operating system.

Please review the IT controls under the AWS Shared Responsibility Model:

Inherited Controls – Controls which a customer fully inherits from AWS.

- Physical and Environmental controls

Shared Controls – Controls which apply to both the infrastructure layer and customer layers, but in completely separate contexts or perspectives. In a shared control, AWS provides the requirements for the infrastructure and the customer must provide their own control implementation within their use of AWS services. Examples include:

- Patch Management – AWS is responsible for patching and fixing flaws within the infrastructure, but customers are responsible for patching their guest OS and applications.
- Configuration Management – AWS maintains the configuration of its infrastructure devices, but a customer is responsible for configuring their own guest operating systems, databases, and applications.
- Awareness & Training – AWS trains AWS employees, but a customer must train their own employees.

Customer Specific – Controls which are solely the responsibility of the customer based on the application they are deploying within AWS services. Examples include:

- Service and Communications Protection or Zone Security which may require a customer to route or zone data within specific security environments.

via - <https://aws.amazon.com/compliance/shared-responsibility-model/>

Encrypting application data - The customers are responsible for encrypting application data.

Configuring IAM Roles - The customers are responsible for configuring IAM Roles.

Reference:

<https://aws.amazon.com/compliance/shared-responsibility-model/>

Domain

Security and Compliance

Question 5Correct

Which security control tool can be used to deny traffic from a specific IP address?

Security Group

Your answer is correct

Network Access Control List (network ACL)

Amazon GuardDuty

VPC Flow Logs

Overall explanation

Correct option:

Network Access Control List (network ACL)

A Network Access Control List (network ACL) is an optional layer of security for your virtual private cloud (VPC) that acts as a firewall for controlling traffic in and out of one or more subnets (i.e. it works at the subnet level). A network ACL has separate inbound and outbound rules, and each rule can either allow or deny traffic.

Network Access Control List (network ACL) Overview:

Network ACL basics

The following are the basic things that you need to know about network ACLs:

- Your VPC automatically comes with a modifiable default network ACL. By default, it allows all inbound and outbound IPv4 traffic and, if applicable, IPv6 traffic.
- You can create a custom network ACL and associate it with a subnet. By default, each custom network ACL denies all inbound and outbound traffic until you add rules.
- Each subnet in your VPC must be associated with a network ACL. If you don't explicitly associate a subnet with a network ACL, the subnet is automatically associated with the default network ACL.
- You can associate a network ACL with multiple subnets. However, a subnet can be associated with only one network ACL at a time. When you associate a network ACL with a subnet, the previous association is removed.
- A network ACL contains a numbered list of rules. We evaluate the rules in order, starting with the lowest numbered rule, to determine whether traffic is allowed in or out of any subnet associated with the network ACL. The highest number that you can use for a rule is 32766. We recommend that you start by creating rules in increments (for example, increments of 10 or 100) so that you can insert new rules where you need to later on.
- A network ACL has separate inbound and outbound rules, and each rule can either allow or deny traffic.
- Network ACLs are stateless, which means that responses to allowed inbound traffic are subject to the rules for outbound traffic (and vice versa).

There are quotas (limits) for the number of network ACLs per VPC, and the number of rules per network ACL. For more information, see [Amazon VPC quotas](#).

via - <https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html>

Incorrect options:

Security Group - A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. Security groups act at the instance level, not at the subnet level. You can specify allow rules, but not deny rules. You can specify separate rules for inbound and outbound traffic.

Amazon GuardDuty - Amazon GuardDuty is a threat detection service that continuously monitors for malicious or unauthorized behavior to help you protect your AWS accounts and workloads. It monitors for activity such as unusual API calls or potentially unauthorized deployments that indicate a possible account compromise. Amazon GuardDuty also detects potentially compromised instances or reconnaissance by attackers. It cannot deny traffic from a specific IP address.

VPC Flow Logs - VPC Flow Logs is a feature that enables you to capture information about the IP traffic going to and from network interfaces in your VPC. Flow log data can be published to Amazon CloudWatch Logs or Amazon Simple Storage Service (Amazon S3). After you've created a flow log, you can retrieve and view its data in the chosen destination. However, it cannot deny traffic from a specific IP address.

Reference:

<https://docs.aws.amazon.com/vpc/latest/userguide/vpc-network-acls.html>

Domain

Security and Compliance

Question 6Correct

Which Amazon Elastic Compute Cloud (Amazon EC2) Auto Scaling feature can help with fault tolerance?

Your answer is correct

Replacing unhealthy Amazon EC2 instances

Having the right amount of computing capacity

Distributing load to Amazon EC2 instances

Lower cost by adjusting the number of Amazon EC2 instances

Overall explanation

Correct option:

Replacing unhealthy Amazon EC2 instances

Amazon EC2 Auto Scaling helps you maintain application availability and allows you to automatically add or remove Amazon EC2 instances according to the conditions you define. You can use the fleet management features of Amazon EC2 Auto Scaling

to maintain the health and availability of your fleet. You can also use the dynamic and predictive scaling features of Amazon EC2 Auto Scaling to add or remove EC2 instances.

Amazon EC2 Auto Scaling can detect when an instance is unhealthy, terminate it, and replace it with a new one.

Incorrect options:

Lower cost by adjusting the number of Amazon EC2 instances - Amazon EC2 Auto Scaling adds instances only when needed, and can scale across purchase options to optimize performance and cost. However, this will not help with fault tolerance.

Distributing load to Amazon EC2 instances - Even though this helps with fault tolerance and is often used with Amazon EC2 Auto Scaling, it is a feature of Elastic Load Balancing (ELB) and not an Amazon EC2 Auto Scaling. Elastic Load Balancing automatically distributes incoming application traffic across multiple targets, such as Amazon EC2 instances, containers, IP addresses, and Lambda functions. It can handle the varying load of your application traffic in a single Availability Zone (AZ) or across multiple Availability Zones (AZs).

Having the right amount of computing capacity - Amazon EC2 Auto Scaling ensures that your application always has the right amount of computing capacity, so your application can handle the workload.

Reference:

<https://aws.amazon.com/ec2/autoscaling/>

Domain

Cloud Concepts

Question 7Correct

What is the primary use case for Amazon GuardDuty?

Your answer is correct

Detecting malicious activity and threats in your AWS accounts and workloads

Enforcing secure communication between VPCs using network traffic filtering

Protecting web applications from common exploits and vulnerabilities such as SQL injection

Encrypting data in transit between AWS services using TLS certificates.

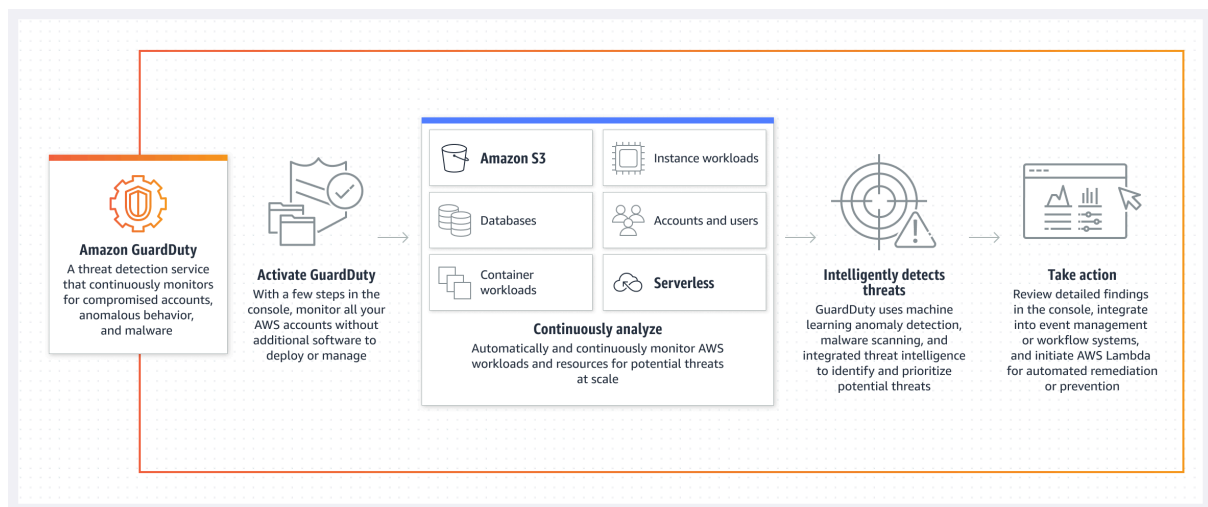
Overall explanation

Correct option:

Detecting malicious activity and threats in your AWS accounts and workloads

Amazon GuardDuty is a managed threat detection service that identifies potential security risks by analyzing logs, such as AWS CloudTrail, VPC Flow Logs, and DNS query logs. It helps organizations detect suspicious or malicious activity in their AWS environments.

Amazon GuardDuty Overview:



via - <https://aws.amazon.com/guardduty/>

Incorrect options:

Enforcing secure communication between VPCs using network traffic filtering - This is a feature provided by AWS Network Firewall or Security Groups, which allow you to control network traffic into and out of your VPCs. GuardDuty detects malicious activities but does not enforce network traffic filtering.

Encrypting data in transit between AWS services using TLS certificates. - AWS Certificate Manager (ACM) or AWS Key Management Service (KMS) is responsible for managing encryption and TLS certificates. GuardDuty does not handle encryption but focuses on identifying potential security threats.

Protecting web applications from common exploits and vulnerabilities such as SQL injection - This is the primary use case for AWS Web Application Firewall (WAF), which protects web applications from vulnerabilities like SQL injection and cross-site scripting. GuardDuty detects threats but does not directly protect applications.

Reference:

<https://aws.amazon.com/guarddduty/>

Domain

Technology

Question 8Correct

A multinational company has just moved its infrastructure to AWS Cloud and has employees traveling to different offices around the world. How should the company set the AWS accounts?

Create an IAM user for each user in each AWS region

Your answer is correct

There is nothing to do, AWS Identity and Access Management (AWS IAM) is a global service

Create global permissions so users can access resources from all around the world

As employees travel, they can use other employees' accounts

Overall explanation

Correct option:

There is nothing to do, AWS Identity and Access Management (AWS IAM) is a global service

AWS Identity and Access Management (IAM) enables you to manage access to AWS services and resources securely. Using IAM, you can create and manage IAM users and IAM user groups, and use permissions to allow and deny their access to AWS resources.

AWS IAM is a global service. Users created within IAM can access their accounts all around the world, and deploy resources in every region.

Incorrect options:

Create an IAM user for each user in each AWS region - IAM users can access their accounts from different AWS regions.

Create global permissions so users can access resources from all around the world - AWS Identity and Access Management (AWS IAM) is a global service. You can use it globally without implementing anything.

As employees travel, they can use other employees' accounts - You should never share your IAM user credentials.

Reference:

<https://aws.amazon.com/iam/>

Domain

Security and Compliance

Question 9Correct

Which AWS service can be used to subscribe to an RSS feed to be notified of the status of all AWS service interruptions?

AWS Health Dashboard - Your Account Health

Your answer is correct

AWS Health Dashboard - Service Health

Amazon Simple Notification Service (Amazon SNS)

AWS Lambda

Overall explanation

Correct option:

AWS Health Dashboard - Service Health

The AWS Health Dashboard – Service health is the single place to learn about the availability and operations of AWS services. You can view the overall status of AWS services, and you can sign in to view personalized communications about your particular AWS account or organization.

You can check on this page <https://health.aws.amazon.com/health/status> to get current status information.

The AWS Health Dashboard – Service health offers the possibility to subscribe to an RSS feed to be notified of interruptions to each service.

Incorrect options:

Amazon Simple Notification Service (Amazon SNS) - Amazon Simple Notification Service (Amazon SNS) is a highly available, durable, secure, fully managed pub/sub messaging service that enables you to decouple microservices, distributed systems, and serverless applications. It can be used to deliver notifications, but it does not provide the current services' status.

AWS Health Dashboard - Your Account Health - Your AWS Health Dashboard – Your Account Health provides alerts and remediation guidance when AWS is experiencing events that may impact you.

AWS Lambda - AWS Lambda lets you run code without provisioning or managing servers. It does not provide all AWS services' status.

Reference:

<https://health.aws.amazon.com/health/status>

Domain

Technology

Question 10Correct

A company would like to separate cost for AWS services by the department for cost allocation. Which of the following is the simplest way to achieve this task?

Your answer is correct

Create tags for each department

Create different virtual private cloud (VPCs) for different departments

Create different accounts for different departments

Create one account for all departments and share this account

Overall explanation

Correct option:

Create tags for each department

You can assign metadata to your AWS resources in the form of tags. Each tag is a label consisting of a user-defined key and value. Tags can help you manage, identify, organize, search for, and filter resources. You can create tags to categorize resources by purpose, owner, environment, or other criteria.

Typically, you use business tags such as cost center/business unit, customer, or project to associate AWS costs with traditional cost-allocation dimensions. But a cost allocation report can include any tag. This lets you associate costs with technical or security dimensions, such as specific applications, environments, or compliance programs.

Example of tagging for cost optimization:

Total Cost ▾	user:Owner ▾	user:Stack ▾	user:Cost Center ▾	user:Application ▾
0.95	DbAdmin	Test	80432	Widget2
0.01	DbAdmin	Test	80432	Widget2
3.84	DbAdmin	Prod	80432	Widget2
6.00	DbAdmin	Test	78925	Widget1
234.63	SysEng	Prod	78925	Widget1
0.73	DbAdmin	Test	78925	Widget1
0.00	DbAdmin	Prod	80432	Portal
2.47	DbAdmin	Prod	78925	Portal

via -

https://docs.aws.amazon.com/general/latest/gr/aws_tagging.html

Incorrect options:

Create different accounts for different departments - Users can belong to several departments. Therefore, having different accounts for different departments would imply some users having several accounts. This is contrary to the security best practice: one physical user = one account. Also, it is much simpler to set up tags for tracking costs for each department.

Create one account for all departments and share this account - Sharing accounts is not a security best practice, and is not recommended.

Create different virtual private cloud (VPCs) for different departments - Creating different VPCs will not help with separating costs.

Reference:

https://docs.aws.amazon.com/general/latest/gr/aws_tagging.html

Domain

Billing and Pricing

Question 11 Correct

Which of the following billing timeframes is applied when running a Windows EC2 on-demand instance?

Pay per minute

Your answer is correct

Pay per second

Pay per hour

Pay per day

Overall explanation

Correct option:

Pay per second

With On-Demand instances, you only pay for the Amazon EC2 instances you use. The use of On-Demand instances frees you from the costs and complexities of planning, purchasing, and maintaining hardware and transforms what are commonly large fixed costs into much smaller variable costs.

When running a Windows EC2 on-demand instance, pay-per-second pricing is applied.

Incorrect options:

Pay per hour - When running an Amazon Windows EC2 On-demand instance, pay-per-second pricing is applied. Windows-based EC2 instances used to follow pay-per-hour pricing earlier.

Pay per minute - Pay per minute pricing is not available for Windows EC2 on-demand instances, or any other type of on-demand EC2 instance.

Pay per day - Pay per day pricing is not available for Windows EC2 on-demand instances, or any other type of on-demand EC2 instance.

Reference:

<https://aws.amazon.com/ec2/pricing/>

Domain

Billing and Pricing

Question 12Correct

Adding more CPU/RAM to an Amazon Elastic Compute Cloud (Amazon EC2) instance represents which of the following?

Loose coupling

Horizontal scaling

Your answer is correct

Vertical scaling

Managing increasing volumes of data

Overall explanation

Correct option:

Vertical scaling

A vertically scalable system is constrained to running its processes on only one computer; in such systems, the only way to increase performance is to add more resources into one computer in the form of faster (or more) CPUs, memory, or storage.

Incorrect options:

Horizontal scaling - A horizontally scalable system is one that can increase capacity by adding more computers to the system.

Managing increasing volumes of data - Traditional data storage and analytics tools can no longer provide the agility and flexibility required to deliver relevant business insights. That's why many organizations are shifting to a data lake architecture. A data lake is an architectural approach that allows you to store massive amounts of data in a central location so that it's readily available to be categorized, processed, analyzed, and consumed by diverse groups within your organization.

Loose coupling - As application complexity increases, a desirable attribute of an IT system is that it can be broken into smaller, loosely coupled components. This means that IT systems should be designed in a way that reduces interdependencies—a change or a failure in one component should not cascade to other components.

Reference:

<https://wa.aws.amazon.com/wat.concept.horizontal-scaling.en.html>

Domain

Cloud Concepts

Question 13Correct

Which of the following statements is INCORRECT regarding Amazon EBS Elastic Volumes?

Your answer is correct

Amazon EBS Elastic Volumes can be bound to several Availability Zones (AZs)

Amazon EBS Elastic Volumes are bound to a specific Availability Zone (AZ)

Amazon EBS Elastic Volumes can persist data after their termination

Amazon EBS Elastic Volumes can be mounted to one instance at a time

Overall explanation

Correct option:

Amazon EBS Elastic Volumes can be bound to several Availability Zones (AZs)

An Amazon EBS Elastic Volume is a durable, block-level storage device that you can attach to your instances. After you attach a volume to an instance, you can use it as you would use a physical hard drive.

When using Amazon EBS Elastic Volumes, the volume, and the instance must be in the same Availability Zone (AZ).

Incorrect options:

Amazon EBS Elastic Volumes can be mounted to one instance at a time - At the Certified Cloud Practitioner level, Amazon EBS Elastic Volumes can be mounted to one instance at a time. It is also possible that an Amazon EBS Elastic Volume is not mounted to an instance.

Amazon EBS Elastic Volumes are bound to a specific Availability Zone (AZ) - As mentioned, when using Amazon EBS Elastic Volumes, the volume and the instance must be in the same Availability Zone(AZ).

Amazon EBS Elastic Volumes can persist data after their termination - Unlike an Amazon EC2 instance store, an Amazon EBS Elastic Volume is off-instance storage that can persist independently from the life of an instance.

Reference:

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ebs-volumes.html>

Domain

Technology

Question 14Correct

Which AWS service can be used to send, store, and receive messages between software components at any volume to decouple application tiers?

AWS Elastic Beanstalk

Your answer is correct

Amazon Simple Queue Service (Amazon SQS)

Amazon Simple Notification Service (Amazon SNS)

AWS Organizations

Overall explanation

Correct option:

Amazon Simple Queue Service (Amazon SQS)

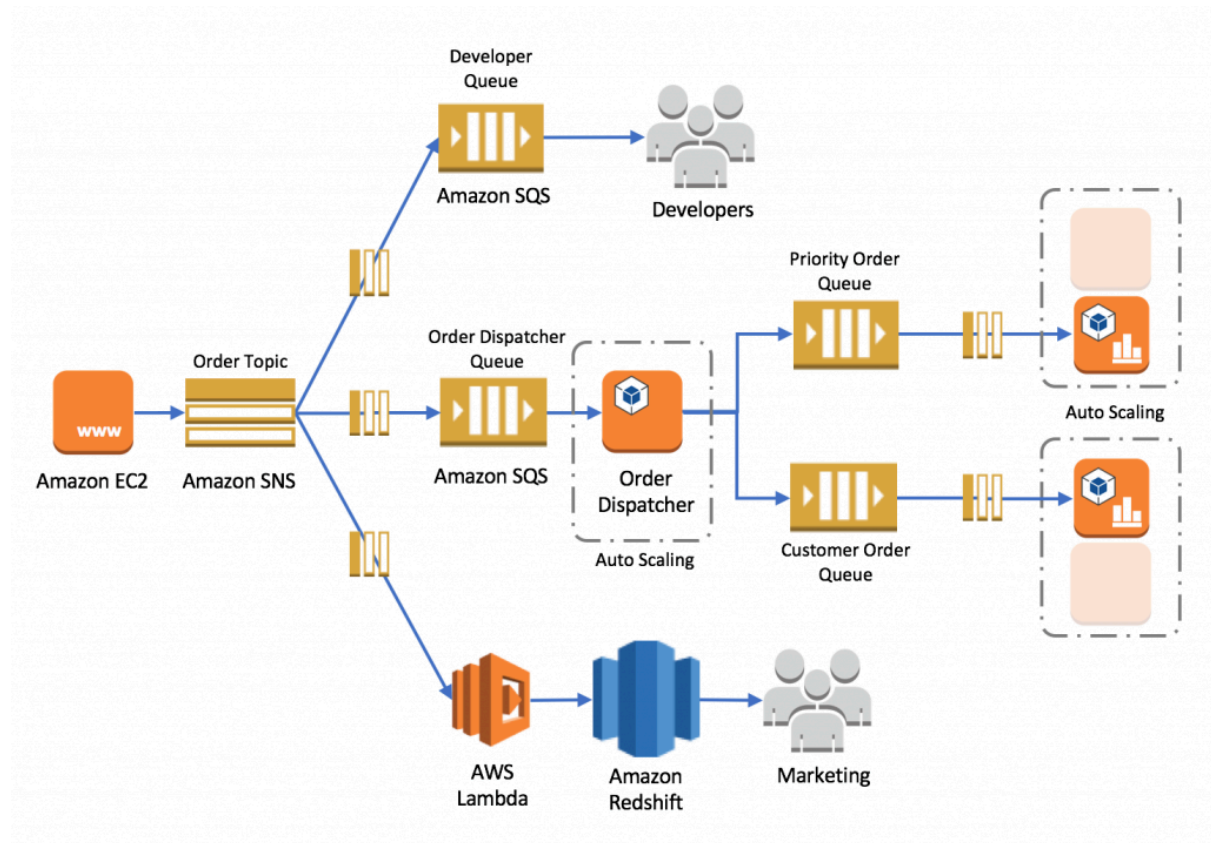
Amazon Simple Queue Service (Amazon SQS) is a fully managed message queuing service that enables you to decouple and scale microservices, distributed systems, and serverless applications. Amazon SQS eliminates the complexity and overhead associated with managing and operating message-oriented middleware and empowers developers to focus on differentiating work.

Using Amazon Simple Queue Service (Amazon SQS), you can send, store, and receive messages between software components at any volume, without losing messages or requiring other services to be available.

Incorrect options:

Amazon Simple Notification Service (Amazon SNS) - Amazon Simple Notification Service (Amazon SNS) is a highly available, durable, secure, fully managed pub/sub messaging service that enables you to decouple microservices, distributed systems, and serverless applications.

Please review this reference architecture for building a decoupled order processing system using Amazon Simple Notification Service (Amazon SNS) and Amazon Simple Queue Service (Amazon SQS):



via -

<https://aws.amazon.com/blogs/compute/building-loosely-coupled-scalable-c-applications-with-amazon-sqs-and-amazon-sns/>

AWS Elastic Beanstalk - AWS Elastic Beanstalk is an easy-to-use service for deploying and scaling web applications and services. You can simply upload your code, and AWS Elastic Beanstalk automatically handles the deployment, from capacity provisioning, load balancing, and auto scaling to application health monitoring. It is not used to send, store, and receive messages between software components.

AWS Organizations - AWS Organizations offers policy-based management for multiple AWS accounts. With AWS Organizations, you can create groups of accounts, automate account creation, and apply and manage policies for those groups. AWS Organizations enables you to centrally manage policies across multiple accounts, without requiring custom scripts and manual processes. It is not used to send, store, and receive messages between software components.

Reference:

<https://aws.amazon.com/sqs/>

Domain

Technology

Question 15Correct

A Cloud Practitioner would like to deploy identical resources across all AWS regions and accounts using templates while estimating costs. Which AWS service can assist with this task?

AWS CodeDeploy

AWS Directory Service for Microsoft Active Directory (AWS Managed Microsoft AD)

Your answer is correct

AWS CloudFormation

Amazon LightSail

Overall explanation

Correct option:

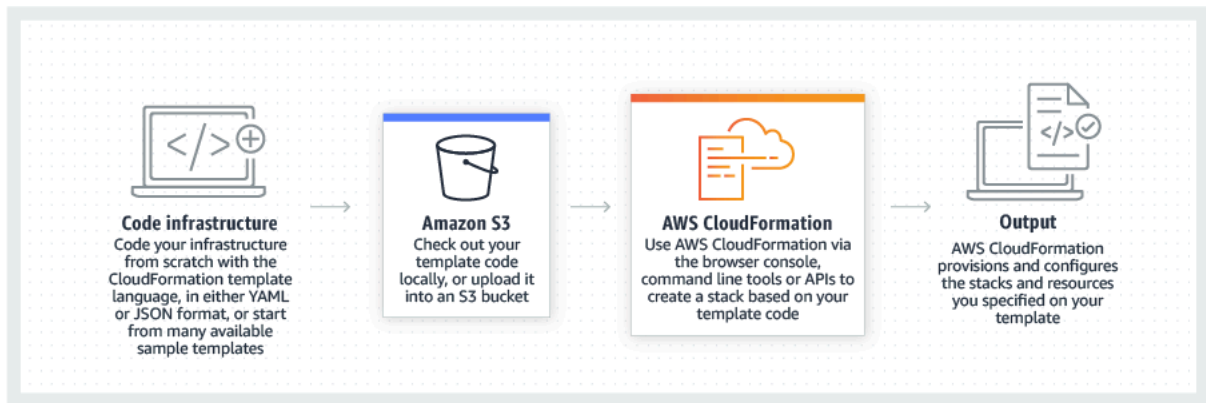
AWS CloudFormation

AWS CloudFormation gives developers and systems administrators an easy way to create and manage a collection of related AWS resources, provisioning and updating them in an orderly and predictable fashion.

You can use the AWS CloudFormation sample templates or create your own templates to describe your AWS resources, and any associated dependencies or runtime parameters, required to run your application. This provides a single source of truth for all your resources and helps you to standardize infrastructure components used across your organization, enabling configuration compliance and faster troubleshooting.

AWS CloudFormation templates allow you to estimate the cost of your resources.

How AWS CloudFormation works:



via - <https://aws.amazon.com/cloudformation/>

Incorrect options:

AWS Directory Service for Microsoft Active Directory (AWS Managed Microsoft AD)

- AWS Directory Service for Microsoft Active Directory (AWS Managed Microsoft AD), also known as AWS Managed Microsoft AD, enables your directory-aware workloads and AWS resources to use managed Active Directory in the AWS Cloud. It is not used to deploy resources.

Amazon Lightsail - Amazon Lightsail is designed to be the easiest way to launch and manage a virtual private server with AWS. It is not best suited when deploying more complex resources, while AWS CloudFormation can.

AWS CodeDeploy - AWS CodeDeploy is a service that automates code deployments to any instance, including Amazon EC2 instances and instances running on-premises. Unlike AWS CloudFormation, it does not deal with infrastructure configuration and orchestration.

Reference:

<https://aws.amazon.com/cloudformation/>

Domain

Technology

Question 16Correct

Which of the following options are the benefits of using AWS Elastic Load Balancing (ELB)? (Select TWO)

Less costly

Agility

Your selection is correct

High availability

Your selection is correct

Fault tolerance

Storage

Overall explanation

Correct options:

High availability

Fault tolerance

Elastic Load Balancing (ELB) automatically distributes incoming application traffic across multiple targets, such as Amazon EC2 instances, containers, and IP addresses. It can handle the varying load of your application traffic in a single Availability Zone (AZ) or across multiple Availability Zones (AZs).

Elastic Load Balancing (ELB) offers three types of load balancers that all feature the high availability, automatic scaling, and robust security necessary to make your applications fault-tolerant: Application Load Balancer (best suited for HTTP and HTTPS traffic), Network Load Balancer (best suited for TCP traffic), and Classic Load Balancer.

Incorrect options:

Agility - Agility refers to new IT resources being only a click away, which means that you reduce the time to make those resources available to your developers from weeks to just minutes. AWS Elastic Load Balancing (ELB) does not help with agility.

Less costly - AWS Elastic Load Balancing (ELB) does not help with reducing costs.

Storage - AWS Elastic Load Balancing (ELB) does not offer storage benefits. It is not a storage-related service.

Reference:

<https://aws.amazon.com/elasticloadbalancing/>

Domain

Cloud Concepts

Question 17Correct

Which of the following statements is the MOST accurate when describing AWS Elastic Beanstalk?

Your answer is correct

It is a Platform as a Service (PaaS) that allows you to deploy and scale web applications and services

It is an Infrastructure as a Service (IaaS) that allows you to deploy and scale web applications and services

It is an Infrastructure as Code (IaC) that allows you to model and provision resources needed for an application

It is a Platform as a Service (PaaS) that allows you to model and provision resources needed for an application

Overall explanation

Correct option:

It is a Platform as a Service (PaaS) that allows you to deploy and scale web applications and services

AWS Elastic Beanstalk makes it even easier for developers to quickly deploy and manage applications in the AWS Cloud. Developers simply upload their applications, and AWS Elastic Beanstalk automatically handles the deployment details of capacity provisioning, load balancing, auto-scaling, and application health monitoring.

It is a Platform as a Service (PaaS) as you only manage the applications and the data.

Please review this overview of the types of Cloud Computing:

Cloud Computing Models

There are three main models for cloud computing. Each model represents a different part of the cloud computing stack.



Infrastructure as a Service (IaaS)

Infrastructure as a Service, sometimes abbreviated as IaaS, contains the basic building blocks for cloud IT and typically provide access to networking features, computers (virtual or on dedicated hardware), and data storage space. Infrastructure as a Service provides you with the highest level of flexibility and management control over your IT resources and is most similar to existing IT resources that many IT departments and developers are familiar with today.



Platform as a Service (PaaS)

Platforms as a service remove the need for organizations to manage the underlying infrastructure (usually hardware and operating systems) and allow you to focus on the deployment and management of your applications. This helps you be more efficient as you don't need to worry about resource procurement, capacity planning, software maintenance, patching, or any of the other undifferentiated heavy lifting involved in running your application.



Software as a Service (SaaS)

Software as a Service provides you with a completed product that is run and managed by the service provider. In most cases, people referring to Software as a Service are referring to end-user applications. With a SaaS offering you do not have to think about how the service is maintained or how the underlying infrastructure is managed; you only need to think about how you will use that particular piece of software. A common example of a SaaS application is web-based email where you can send and receive email without having to manage feature additions to the email product or maintaining the servers and operating systems that the email program is running on.

via - <https://aws.amazon.com/types-of-cloud-computing/>

Incorrect options:

It is an Infrastructure as Code (IaC) that allows you to model and provision resources needed for an application - This is the definition of AWS CloudFormation. AWS CloudFormation gives developers and systems administrators an easy way to create and manage a collection of related AWS resources, provisioning and updating them in an orderly and predictable fashion. You can use the AWS CloudFormation sample templates or create your own templates to describe your AWS resources, and any associated dependencies or runtime parameters, required to run your application.

It is a Platform as a Service (PaaS) that allows you to model and provision resources needed for an application - AWS Elastic Beanstalk is a Platform as a Service (PaaS). However, the service that allows you to model and provision resources needed for an application is AWS CloudFormation.

It is an Infrastructure as a Service (IaaS) that allows you to deploy and scale web applications and services - AWS Elastic Beanstalk allows you to deploy and scale web applications and services, but it is not an Infrastructure as a Service (IaaS). With AWS Elastic Beanstalk, you do not manage the runtime, the middleware, and the operating system.

Reference:

<https://aws.amazon.com/elasticbeanstalk/>

Domain

Technology

Question 18Correct

A media company wants to enable customized content suggestions for the users of its movie streaming platform. Which AWS service can provide these personalized recommendations based on historic data?

Amazon SageMaker

Your answer is correct

Amazon Personalize

Amazon Customize

Amazon Comprehend

Overall explanation

Correct option:

Amazon Personalize

Amazon Personalize enables developers to build applications with the same machine learning (ML) technology used by Amazon.com for real-time personalized recommendations. Amazon Personalize can be used to personalize the end-user experience over any digital channel. Examples include product recommendations for e-commerce, news articles and content recommendation for publishing, media, and social networks, hotel recommendations for travel websites, credit card recommendations for banks, and match recommendations for dating sites. These recommendations and personalized experiences can be delivered over websites, mobile apps, or email/messaging. Amazon Personalize can also be used to customize the user experience when user interaction is over a physical channel, e.g., a meal delivery company could personalize weekly meals to users in a subscription plan.

Amazon Personalize supports the following key use cases:

1. Personalized recommendations
2. Similar items
3. Personalized reranking i.e. rerank a list of items for a user
4. Personalized promotions/notifications

Incorrect options:

Amazon SageMaker - Amazon SageMaker is a fully managed service that provides every developer and data scientist with the ability to build, train, and deploy machine learning (ML) models quickly. Amazon SageMaker removes the heavy lifting from each step of the machine learning process to make it easier to develop high-quality models.

Amazon Customize - There is no such service as Amazon Customize. This option has been added as a distractor.

Amazon Comprehend - Amazon Comprehend is a natural-language processing (NLP) service that uses machine learning to uncover information in unstructured data. Instead of combing through documents, the process is simplified and unseen information is easier to understand.

The service can identify critical elements in data, including references to language, people, and places, and the text files can be categorized by relevant topics. In real-time, you can automatically and accurately detect customer sentiment in your content.

Reference:

<https://aws.amazon.com/personalize/>

Domain

Technology

Question 19Correct

According to the AWS Well-Architected Framework, which of the following statements are recommendations in the Operational Excellence pillar? (Select two)

Your selection is correct

Make frequent, small, reversible changes

Your selection is correct

Anticipate failure

Enable traceability

Use serverless architectures

Automatically recover from failure

Overall explanation

Correct options:

Anticipate failure

Make frequent, small, reversible changes

The Operational Excellence pillar includes the ability to run and monitor systems to deliver business value and to continually improve supporting processes and procedures.

Perform “pre-mortem” exercises to identify potential sources of failure so that they can be removed or mitigated. Test your failure scenarios and validate your understanding of their impact. Test your response procedures to ensure that they are effective, and that teams are familiar with their execution. Set up regular game days to test workloads and team responses to simulated events.

Design workloads to allow components to be updated regularly. Make changes in small increments that can be reversed if they fail (without affecting customers when possible).

The AWS Well-Architected Framework helps you understand the pros and cons of the decisions you make while building systems on AWS. By using the Framework you will learn architectural best practices for designing and operating reliable, secure, efficient, and cost-effective systems in the cloud. It provides a way for you to consistently measure your architectures against best practices and identify areas for improvement.

The AWS Well-Architected Framework is based on six pillars — Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, and Sustainability.

Overview of the six pillars of the AWS Well-Architected Framework:

AWS Well-Architected and the Six Pillars

Framework Overview

The AWS Well-Architected Framework describes key concepts, design principles, and architectural best practices for designing and running workloads in the cloud. By answering a few foundational questions, learn how well your architecture aligns with cloud best practices and gain guidance for making improvements.

[HTML](#) | [Kindle](#) | [Labs](#)



Operational Excellence Pillar

The operational excellence pillar focuses on running and monitoring systems, and continually improving processes and procedures. Key topics include automating changes, responding to events, and defining standards to manage daily operations.

[HTML](#) | [Kindle](#) | [Labs](#)

Security Pillar

The security pillar focuses on protecting information and systems. Key topics include confidentiality and integrity of data, managing user permissions, and establishing controls to detect security events.

[HTML](#) | [Kindle](#) | [Labs](#)

Reliability Pillar

The reliability pillar focuses on workloads performing their intended functions and how to recover quickly from failure to meet demands. Key topics include distributed system design, recovery planning, and adapting to changing requirements.

[HTML](#) | [Kindle](#) | [Labs](#)

Performance Efficiency Pillar

The performance efficiency pillar focuses on structured and streamlined allocation of IT and computing resources. Key topics include selecting resource types and sizes optimized for workload requirements, monitoring performance, and maintaining efficiency as business needs evolve.

Cost Optimization Pillar

The cost optimization pillar focuses on avoiding unnecessary costs. Key topics include understanding spending over time and controlling fund allocation, selecting resources of the right type and quantity, and scaling to meet business needs without overspending.

Sustainability Pillar

The sustainability pillar focuses on minimizing the environmental impacts of running cloud workloads. Key topics include a shared responsibility model for sustainability, understanding impact, and maximizing utilization to minimize required resources and reduce downstream impacts.

via - <https://aws.amazon.com/architecture/well-architected/>

Incorrect options:

Enable traceability - Monitor, alert, and audit actions and changes to your environment in real-time. Integrate logs and metrics with systems to automatically respond and take action. It is a design principle of the Security pillar.

Automatically recover from failure - By monitoring a system for key performance indicators (KPIs), you can trigger automation when a threshold is breached. This allows for automatic notification and tracking of failures, and for automated recovery processes that work around or repair the failure. With more sophisticated automation, it's possible to anticipate and remediate failures before they occur. It is a design principle of the Reliability pillar.

Use serverless architectures - In the cloud, serverless architectures remove the need for you to run and maintain servers to carry out traditional compute activities. For example, storage services can act as static websites, removing the need for web servers, and event services can host your code for you. This not only removes the operational burden of managing these servers but also can lower transactional costs because these managed services operate at a cloud scale. It is a design principle of the Performance Efficiency pillar.

Reference:

<https://wa.aws.amazon.com/index.en.html>

Domain

Cloud Concepts

Question 20Correct

According to the AWS Shared Responsibility Model, which of the following is both the responsibility of AWS and the customer? (Select two)

Disposal of disk drives

Your selection is correct

Configuration management

Customer data

Data center security

Your selection is correct

Operating system (OS) configuration

Overall explanation

Correct options:

Configuration management

Shared Controls – Controls that apply to both the infrastructure layer and customer layers, but in completely separate contexts or perspectives. In a shared control, AWS provides the requirements for the infrastructure and the customer must provide their own control implementation within their use of AWS services.

Configuration Management – AWS maintains the configuration of its infrastructure devices, but a customer is responsible for configuring their own guest operating systems, databases, and applications.

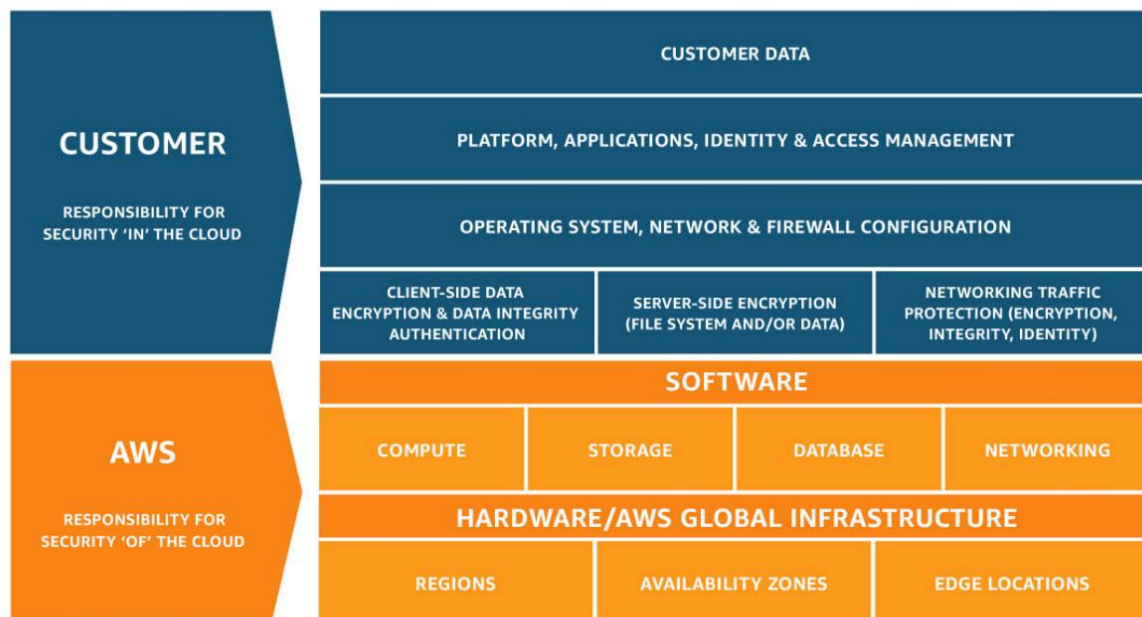
Operating system (OS) configuration

The customers are responsible for "Security IN the cloud". It includes customer data, as well as the guest operating system configuration.

Operating System configuration as a whole is an AWS shared responsibility, but be careful: the host operating system configuration is the responsibility of AWS, and the guest operating system configuration is the responsibility of the customer.

Exam Alert:

Please review the AWS Shared Responsibility Model in detail as you can expect multiple questions on the shared responsibility model in the exam:



via - <https://aws.amazon.com/compliance/shared-responsibility-model/>

Incorrect options:

Customer data

Data center security

Disposal of disk drives

AWS is responsible for "Security OF the cloud". It includes the infrastructure, which is composed of the hardware, software, networking, and facilities that run AWS Cloud services. It includes the disposal and the replacement of disk drives as well as data center security.

Reference:

<https://aws.amazon.com/compliance/shared-responsibility-model/>

Domain

Security and Compliance

Question 21 Correct

Which types of monitoring can be provided by Amazon CloudWatch? (Select TWO)

Your selection is correct

Application performance

Performance and availability of AWS services

API access

Account management

Your selection is correct

Resource utilization

Overall explanation

Correct options:

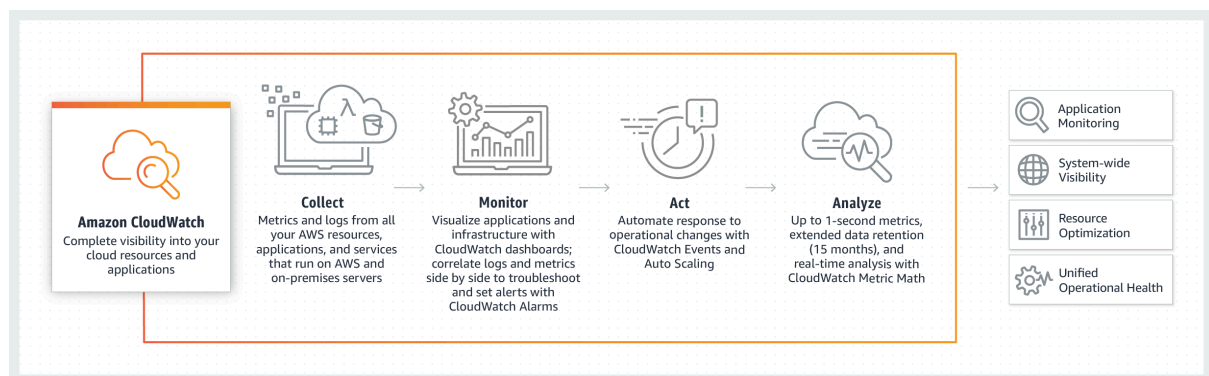
Application performance

Resource utilization

Amazon CloudWatch is a monitoring service for AWS cloud resources and the applications you run on AWS. You can use Amazon CloudWatch to collect and track metrics, collect and monitor log files, and set alarms. Amazon CloudWatch can monitor AWS resources such as Amazon Elastic Compute Cloud (Amazon EC2) instances, Amazon DynamoDB tables, and Amazon Amazon Relational Database Service (Amazon RDS) DB instances, as well as custom metrics generated by your applications and services, and any log files your applications generate.

You can use Amazon CloudWatch to gain system-wide visibility into resource utilization, application performance, and operational health. You can use these insights to react and keep your application running smoothly.

How Amazon CloudWatch works:



via - <https://aws.amazon.com/cloudwatch/>

Incorrect options:

API access - Recording API calls is a feature of AWS CloudTrail, not Amazon CloudWatch.

Performance and availability of AWS services - The AWS Health - Your Account Health Dashboard gives you a personalized view of the performance and availability of the AWS services underlying your AWS resources, not Amazon CloudWatch.

Account management - AWS Identity and Access Management (AWS IAM) is usually used to manage accounts, not Amazon CloudWatch.

References:

<https://aws.amazon.com/cloudwatch/features/>

<https://aws.amazon.com/cloudwatch/>

Domain

Technology

Question 22Correct

A start-up would like to quickly deploy a popular technology on AWS. As a Cloud Practitioner, which AWS tool would you use for this task?

AWS Whitepapers

AWS CodeDeploy

AWS Forums

Your answer is correct

AWS Partner Solutions (formerly Quick Starts)

Overall explanation

Correct option:

AWS Partner Solutions (formerly Quick Starts)

AWS Partner Solutions are automated reference deployments built by Amazon Web Services (AWS) solutions architects and AWS Partners. Partner Solutions help you deploy popular technologies to AWS according to AWS best practices. You can reduce hundreds of manual procedures to a few steps and start using your environment within minutes.

AWS Partner Solutions are automated reference deployments for key workloads on the AWS Cloud. Each Partner Solution launches, configures, and runs the AWS compute, network, storage, and other services required to deploy a specific workload on AWS, using AWS best practices for security and availability.

Incorrect options:

AWS Forums - AWS Forums is an AWS community platform where people can help each other. It is not used to deploy technologies on AWS.

AWS CodeDeploy - AWS CodeDeploy is a service that automates code deployments to any instance, including Amazon EC2 instances and instances running on-premises. It is not suited to rapidly deploy popular technologies on AWS ready to be used immediately.

AWS Whitepapers - AWS Whitepapers are technical content authored by AWS and the AWS community to expand your knowledge of the cloud. They include technical whitepapers, technical guides, reference material, and reference architecture diagrams. You can find useful content for your deployment, but it is not a service that will deploy technologies.

Reference:

<https://aws.amazon.com/quickstart/>

Domain

Technology

Question 23Correct

An engineering team is new to the AWS Cloud and it would like to launch a dev/test environment with low monthly pricing. Which AWS service can address this use case?

Amazon Elastic Container Service (Amazon ECS)

Amazon Elastic Compute Cloud (Amazon EC2)

Your answer is correct

Amazon LightSail

AWS CloudFormation

Overall explanation

Correct option:

Amazon LightSail

Amazon Lightsail is designed to be the easiest way to launch and manage a virtual private server (VPS) with AWS. Amazon Lightsail plans include everything you need to jumpstart your project – a virtual machine, SSD-based storage, data transfer, Domain Name System (DNS) management, and a static IP address – for a low, predictable price.

It is great for people with little cloud experience to launch quickly a popular IT solution ready to use immediately.

Incorrect options:

AWS CloudFormation - AWS CloudFormation gives developers and systems administrators an easy way to create and manage a collection of related AWS resources, provisioning and updating them in an orderly and predictable fashion. Using AWS CloudFormation requires experience as resources are deployed within a virtual private cloud (VPC).

Amazon Elastic Compute Cloud (Amazon EC2) - Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides secure, resizable compute capacity in the cloud. It is designed to make web-scale computing easier for developers. Deploying a dev/test environment with Amazon EC2 requires experience as instances are deployed within a virtual private cloud (VPC).

Amazon Elastic Container Service (Amazon ECS) - Amazon Elastic Container Service (Amazon ECS) is a highly scalable, high-performance container orchestration service that supports Docker containers and allows you to easily run and scale containerized applications on AWS. Amazon ECS eliminates the need for you to install and operate your own container orchestration software, manage and scale a cluster of virtual machines, or schedule containers on those virtual machines. Using Amazon ECS requires experience.

Reference:

<https://aws.amazon.com/lightsail/>

Domain

Technology

Question 24Incorrect

Which of the following options is NOT a feature of Amazon Inspector?

Analyze against unintended network accessibility

Your answer is incorrect

Automate security assessments

Correct answer

Track configuration changes

Inspect running operating systems (OS) against known vulnerabilities

Overall explanation

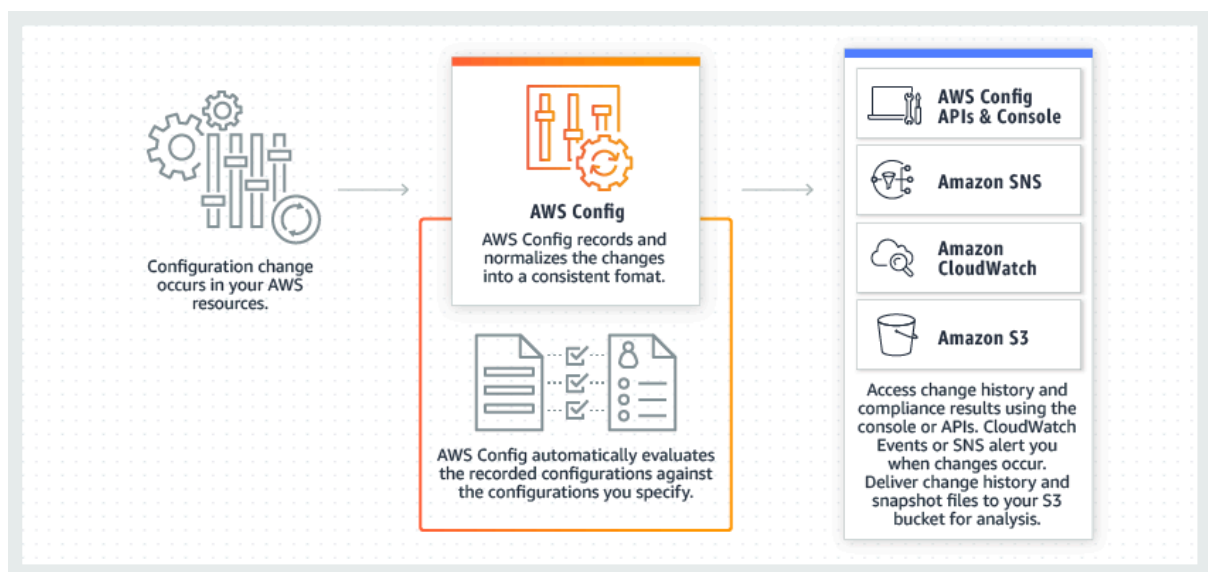
Correct option:

Track configuration changes

Tracking configuration changes is a feature of AWS Config.

AWS Config is a service that enables you to assess, audit, and evaluate the configurations of your AWS resources. Config continuously monitors and records your AWS resource configurations and allows you to automate the evaluation of recorded configurations against desired configurations.

How AWS Config works:



via - <https://aws.amazon.com/config/>

Incorrect options:

Automate security assessments

Analyze against unintended network accessibility

Inspect running operating systems (OS) against known vulnerabilities

These options are all features of Amazon Inspector.

Amazon Inspector is an automated security assessment service that helps improve the security and compliance of applications deployed on AWS. Amazon Inspector automatically assesses applications for exposure, vulnerabilities, and deviations from best practices.

Amazon Inspector security assessments help you check for unintended network accessibility of your Amazon EC2 instances and for vulnerabilities on those EC2 instances.

Amazon Inspector also offers predefined software called an agent that you can optionally install in the operating system of the EC2 instances that you want to assess. The agent monitors the behavior of the EC2 instances, including network, file system, and process activity. It also collects a wide set of behavior and configuration data (telemetry).

References:

<https://aws.amazon.com/config/>

<https://aws.amazon.com/inspector/>

Domain

Security and Compliance

Question 25Correct

A research lab needs to be notified in case of a configuration change for security and compliance reasons. Which AWS service can assist with this task?

AWS Trusted Advisor

Amazon Inspector

Your answer is correct

AWS Config

AWS Secrets Manager

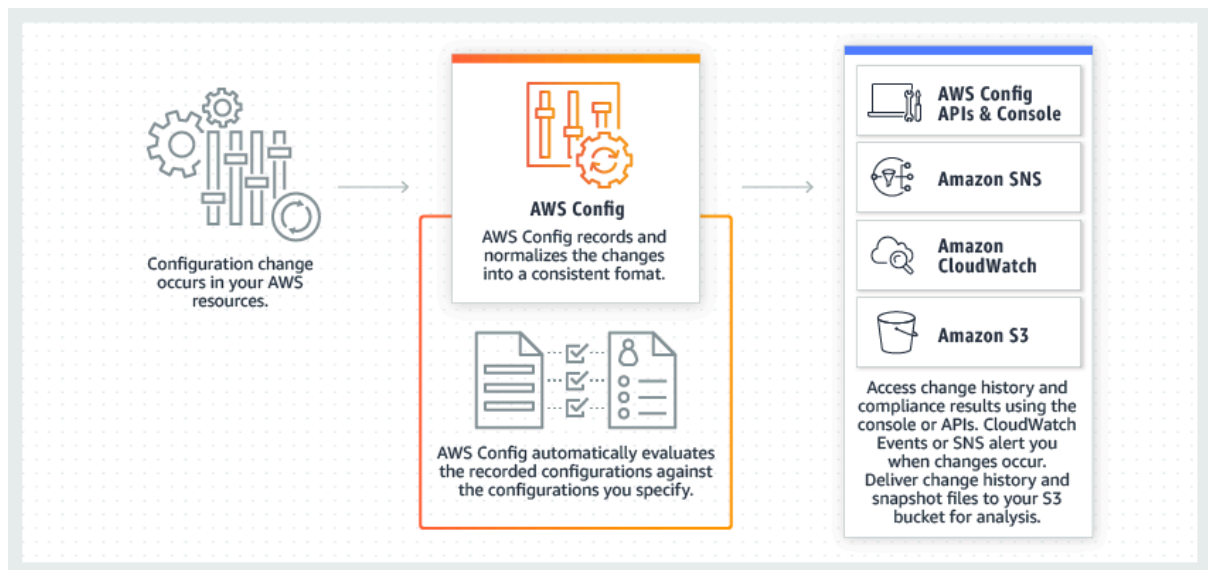
Overall explanation

Correct option:

AWS Config

AWS Config is a service that enables you to assess, audit, and evaluate the configurations of your AWS resources. AWS Config continuously monitors and records your AWS resource configurations and allows you to automate the evaluation of recorded configurations against desired configurations. With AWS Config, you can review changes in configurations and relationships between AWS resources, dive into detailed resource configuration histories, and determine your overall compliance against the configurations specified in your internal guidelines. This enables you to simplify compliance auditing, security analysis, change management, and operational troubleshooting.

How AWS Config works:



via - <https://aws.amazon.com/config/>

Incorrect options:

Amazon Inspector - Amazon Inspector is an automated security assessment service that helps improve the security and compliance of applications deployed on your Amazon EC2 instances. Amazon Inspector automatically assesses applications for exposure, vulnerabilities, and deviations from best practices. It cannot notify configuration changes.

AWS Trusted Advisor - AWS Trusted Advisor is an online resource to help you reduce cost, increase performance, and improve security by optimizing your AWS environment. Trusted Advisor provides real-time guidance to help you provision your resources following AWS best practices. It cannot notify configuration changes.

AWS Secrets Manager - AWS Secrets Manager helps you protect the secrets needed to access your applications, services, and IT resources. The service enables you to easily rotate, manage, and retrieve database credentials, API keys, and other secrets throughout their lifecycle. It cannot notify configuration changes.

Reference:

<https://aws.amazon.com/config/>

Domain

Security and Compliance

Question 26Correct

An enterprise is planning to move one of its older applications from its local data center to AWS. The IT team wants the fastest migration path and has decided not to update the application code or make any architectural changes.

Which migration strategy is the most appropriate for this scenario?

Replatform

Refactor

Repurchase

Your answer is correct

Rehost

Overall explanation

Correct option:

Rehost

About the migration strategies

↓ PDF

↓ RSS

☐ Focus mode

A *migration strategy* is the approach used to migrate a workload into the AWS Cloud. There are seven migration strategies for moving applications to the cloud, known as the 7 Rs:

- [Retire](#)
- [Retain](#)
- [Rehost](#)
- [Relocate](#)
- [Repurchase](#)
- [Replatform](#)
- [Refactor or re-architect](#)

Common strategies for large migrations include rehost, replatform, relocate, and retire. Refactor is not recommended for large migrations because it involves modernizing the application during the migration. This is the most complex of the migration strategies, and it can be complicated to manage for a large number of applications. Instead, we recommend rehosting, relocating, or replatforming the application and then modernizing the application after the migration is complete.

Selecting migration strategies is critical to a large migration. You might have selected migration strategies in the mobilize phase or during the initial portfolio assessment. This section reviews each migration strategy and their common use cases.

via -

<https://docs.aws.amazon.com/prescriptive-guidance/latest/large-migration-guide/migration-strategies.html>

Summary of migration strategies:

Strategy	Also Known As	Description	Use Case	Effort Level
Rehost	Lift and Shift	Move applications to the cloud without changing the architecture.	Quick migration of legacy systems to AWS with minimal disruption.	Low
Replatform	Lift, Tinker, and Shift	Move to the cloud with small optimizations (e.g., moving a self-managed DB to Amazon RDS).	Reduce management effort without significant architectural change.	Low to Medium
Repurchase	Drop and Shop	Replace an existing application with a SaaS alternative.	Move from proprietary solutions to cloud-native SaaS products.	Medium
Refactor	Re-architect	Redesign the application to leverage cloud-native features.	Modernize applications for scalability, agility, or performance.	High
Retire	Decommission	Shut down applications no longer needed.	Remove outdated or redundant systems.	Low
Retain	Revisit	Keep certain applications on-premises (e.g., due to compliance).	Postpone migration for systems not ready for cloud.	Low
Relocate	Hypervisor-level Migration	Migrate large numbers of workloads without modifying them using tools like CloudEndure.	Data center evacuation or VMWare Cloud on AWS migrations.	Medium

Rehosting, commonly referred to as “lift and shift,” involves moving applications to the cloud as-is, without making any changes to the application code or architecture. This is typically the fastest and least disruptive migration path, especially suitable for legacy workloads that are stable but not cloud-optimized.

Incorrect options:

Repurchase - Repurchasing, also known as “drop and shop,” involves abandoning the existing application and switching to a new cloud-based solution, often a SaaS product. For example, moving from an on-premises CRM to Salesforce. This strategy often requires adapting business processes to fit the new system. Since the scenario specifies no changes to the application, this strategy is not applicable.

Replatform - Replatforming, or “lift, tinker, and shift,” involves moving the application to the cloud with minor modifications to optimize for cloud performance (e.g., migrating from a self-hosted MySQL database to Amazon RDS). While it preserves most of the architecture, some changes are still made. Since the question specifies that no changes are to be made, replatforming is not the best fit.

Refactor - Refactoring (also called re-architecting) involves modifying or re-writing parts of the application to take advantage of cloud-native features like serverless architecture, microservices, or containers. This option requires the most effort and time. Since the company in the scenario wants no changes, refactoring is not suitable.

Reference:

<https://docs.aws.amazon.com/prescriptive-guidance/latest/large-migration-guide/migration-strategies.html>

Domain

Cloud Concepts

Question 27Correct

Which of the following statements is an AWS best practice when architecting for the Cloud?

Servers, not services

Your answer is correct

Automation

Close coupling

Security comes last

Overall explanation

Correct option:

Automation

Automation should be implemented to improve both your system's stability and the efficiency of your organization. There are many services to automate application architecture (AWS Elastic Beanstalk, Auto Scaling, AWS Lambda, etc.) to ensure more resiliency, scalability, and performance.

Incorrect options:

Servers, not services - The correct best practice is: "Services, not servers". AWS recommends developing, managing, and operating applications, especially at scale, using the broad set of compute, storage, database, analytics, applications, and deployment services offered by AWS to move faster and lower IT costs.

Close coupling - The correct best practice is: "Loose coupling". AWS recommends that, as application complexity increases, IT systems should be designed in a way that reduces interdependencies. Therefore, a change or a failure in one component should not cascade to other components.

Security comes last - AWS allows you to improve your security in many, more simple ways. Therefore, you should take advantage of this and implement a high level of security.

Reference:

<https://aws.amazon.com/architecture/well-architected/>

Domain

Cloud Concepts

Question 28Correct

A company would like to optimize Amazon Elastic Compute Cloud (Amazon EC2) costs. Which of the following actions can help with this task? (Select TWO)

Build its own servers

Vertically scale the EC2 instances

Your selection is correct

Purchase Amazon EC2 Reserved instances (RIs)

Opt for a higher AWS Support plan

Your selection is correct

Set up Auto Scaling groups to align the number of instances with the demand

Overall explanation

Correct options:

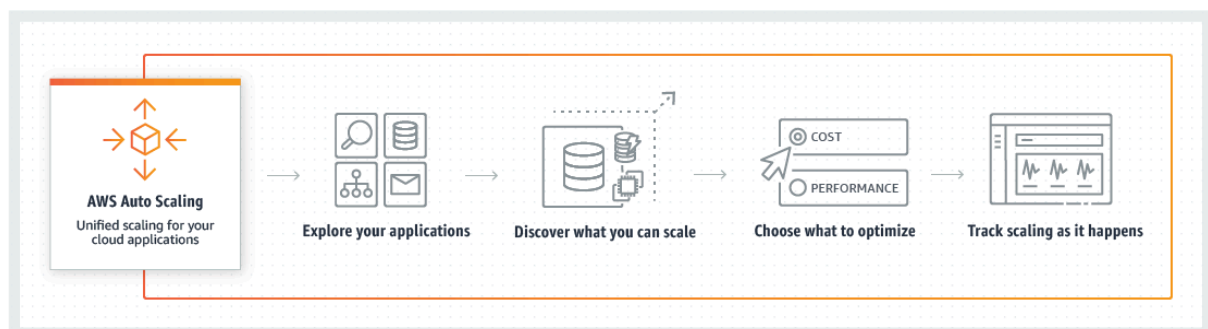
Set up Auto Scaling groups to align the number of instances with the demand

Purchase Amazon EC2 Reserved instances (RIs)

An Auto Scaling group contains a collection of Amazon EC2 instances that are treated as a logical grouping for automatic scaling and management. You can adjust its size to meet demand, either manually or by using automatic scaling.

AWS Auto Scaling can help you optimize your utilization and cost efficiencies when consuming AWS services so you only pay for the resources you need.

How AWS Auto Scaling works:



via - <https://aws.amazon.com/autoscaling/>

Amazon EC2 Reserved Instances (RI) provide a significant discount (up to 72%) compared to On-Demand pricing and provide a capacity reservation when used in a specific Availability Zone (AZ).

EC2 Pricing Options Overview:

On-Demand

With On-Demand instances, you pay for compute capacity by the hour or the second depending on which instances you run. No longer-term commitments or upfront payments are needed. You can increase or decrease your compute capacity depending on the demands of your application and only pay the specified per hourly rates for the instance you use.

On-Demand instances are recommended for:

- Users that prefer the low cost and flexibility of Amazon EC2 without any up-front payment or long-term commitment
- Applications with short-term, spiky, or unpredictable workloads that cannot be interrupted
- Applications being developed or tested on Amazon EC2 for the first time

[See On-Demand pricing »](#)

Spot instances

Amazon EC2 Spot instances allow you to request spare Amazon EC2 computing capacity for up to 90% off the On-Demand price. [Learn More.](#)

Spot instances are recommended for:

- Applications that have flexible start and end times
- Applications that are only feasible at very low compute prices
- Users with urgent computing needs for large amounts of additional capacity

[See Spot pricing »](#)

Savings Plans

Savings Plans are a flexible pricing model that offer low prices on EC2 and Fargate usage, in exchange for a commitment to a consistent amount of usage (measured in \$/hour) for a 1 or 3 year term.

Dedicated Hosts

A Dedicated Host is a physical EC2 server dedicated for your use. Dedicated Hosts can help you reduce costs by allowing you to use your existing server-bound software licenses, including Windows Server, SQL Server, and SUSE Linux Enterprise Server (subject to your license terms), and can also help you meet compliance requirements. [Learn more.](#)

- Can be purchased On-Demand (hourly).
- Can be purchased as a Reservation for up to 70% off the On-Demand price.

[See Dedicated pricing »](#)

Reserved Instances

Reserved Instances provide you with a significant discount (up to 75%) compared to On-Demand instance pricing. In addition, when Reserved Instances are assigned to a specific Availability Zone, they provide a capacity reservation, giving you additional confidence in your ability to launch instances when you need them.

For applications that have steady state or predictable usage, Reserved Instances can provide significant savings compared to using On-Demand instances. See [How to Purchase Reserved Instances](#) for more information.

Reserved Instances are recommended for:

- Applications with steady state usage
- Applications that may require reserved capacity
- Customers that can commit to using EC2 over a 1 or 3 year term to reduce their total computing costs

via - <https://aws.amazon.com/ec2/pricing/>

Incorrect options:

Vertically scale the EC2 instances - Vertically scaling EC2 instances (increasing one computer performance by adding CPUs, memory, and storage) is limited and is way more expensive than scaling horizontally (adding more computers to the system).

Opt for a higher AWS Support plan - The AWS Support plans do not help with EC2 costs.

Build its own servers - Building your own servers is more expensive than using EC2 instances in the cloud. You're more likely to spend more money than saving money.

References:

<https://docs.aws.amazon.com/autoscaling/ec2/userguide/AutoScalingGroup.html>

<https://aws.amazon.com/ec2/pricing/reserved-instances/>

<https://wa.aws.amazon.com/wat.concept.horizontal-scaling.en.html>

<https://aws.amazon.com/autoscaling/>

Domain

Billing and Pricing

Question 29Correct

A start-up would like to monitor its cost on the AWS Cloud and would like to choose an optimal Savings Plan. As a Cloud Practitioner, which AWS service would you use?

AWS Budgets

AWS Cost & Usage Report (AWS CUR)

Your answer is correct

AWS Cost Explorer

AWS Pricing Calculator

Overall explanation

Correct option:

AWS Cost Explorer

AWS Cost Explorer lets you explore your AWS costs and usage at both a high level and at a detailed level of analysis and empowers you to dive deeper using several filtering dimensions (e.g., AWS Service, AWS Region, Linked Account, etc.). AWS Cost Explorer also gives you access to a set of default reports to help you get started, while also allowing you to create custom reports from scratch.

Customers can receive Savings Plan recommendations at the member (linked) account level in addition to the existing AWS organization-level recommendations in AWS Cost Explorer.

Incorrect options:

AWS Cost & Usage Report (AWS CUR) - The AWS Cost & Usage Report (AWS CUR) is a single location for accessing comprehensive information about your AWS costs and usage. It does not provide Savings Plan recommendations.

AWS Pricing Calculator - AWS Pricing Calculator lets you explore AWS services, and create an estimate for the cost of your use cases on AWS. It does not provide Savings Plan recommendations.

AWS Budgets - AWS Budgets gives you the ability to set custom budgets that alert you when your costs or usage exceed (or are forecasted to exceed) your budgeted amount. You can also use AWS Budgets to set reserved instance (RI) utilization or

coverage targets and receive alerts when your utilization drops below the threshold you define. It does not provide Savings Plan recommendations.

Exam Alert:

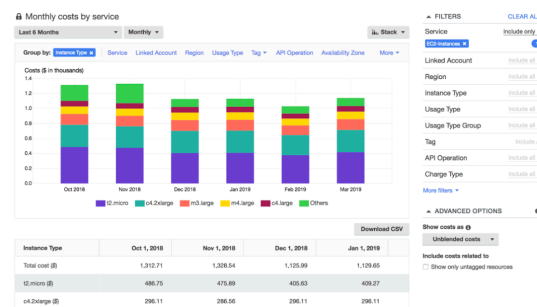
Please review the differences between "AWS Cost & Usage Report (AWS CUR)" and "AWS Cost Explorer". Think of "AWS Cost & Usage Report (AWS CUR)" as a cost management tool providing the most detailed cost and usage data for your AWS account. It can provide reports that break down your costs by the hour into your Amazon Simple Storage Service (Amazon S3) bucket. On the other hand, "AWS Cost Explorer" is more of a high-level cost management tool that helps you visualize the costs and usage associated with your AWS account.

"AWS Cost Explorer" vs "AWS Cost & Usage Report (AWS CUR)":

Monthly Costs by AWS Service

AWS Cost Explorer includes a default report that helps you visualize the costs and usage associated with your top five cost-accruing AWS services, and gives you a detailed breakdown on all services in the table view. The reports let you adjust the time range to view historical data going back up to twelve months to gain an understanding of your cost trends.

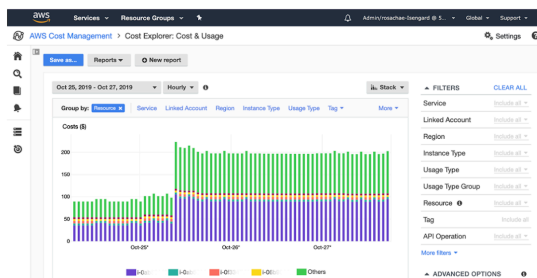
[Launch the Monthly Costs by AWS Service report »](#)



Hourly and Resource Level Granularity

AWS Cost Explorer helps you visualize, understand, and manage your AWS costs and usage over a daily or monthly granularity. The solution also lets you dive deeper using granular filtering and grouping dimensions such as Usage Type and Tags. You can also access your data with further granularity by enabling hourly and resource level granularity.

[Get started using Hourly and Resource Level Granularity »](#)



via - <https://aws.amazon.com/aws-cost-management/aws-cost-explorer/>



via -

<https://aws.amazon.com/aws-cost-management/aws-cost-and-usage-reporting/>

References:

<https://aws.amazon.com/about-aws/whats-new/2020/03/aws-cost-explorer-now-offers-savings-plans-recommendations-for-member-linked-accounts/>

<https://docs.aws.amazon.com/cur/latest/userguide/what-is-cur.html>

<https://aws.amazon.com/aws-cost-management/aws-cost-explorer/>

<https://aws.amazon.com/aws-cost-management/aws-cost-and-usage-reporting/>

Domain

Billing and Pricing

Question 30Correct

A company would like to create a private, high bandwidth network connection between its on-premises data centers and AWS Cloud. As a Cloud Practitioner, which of the following options would you recommend?

Your answer is correct

AWS Direct Connect

VPC Endpoints

VPC peering connection

AWS Site-to-Site VPN

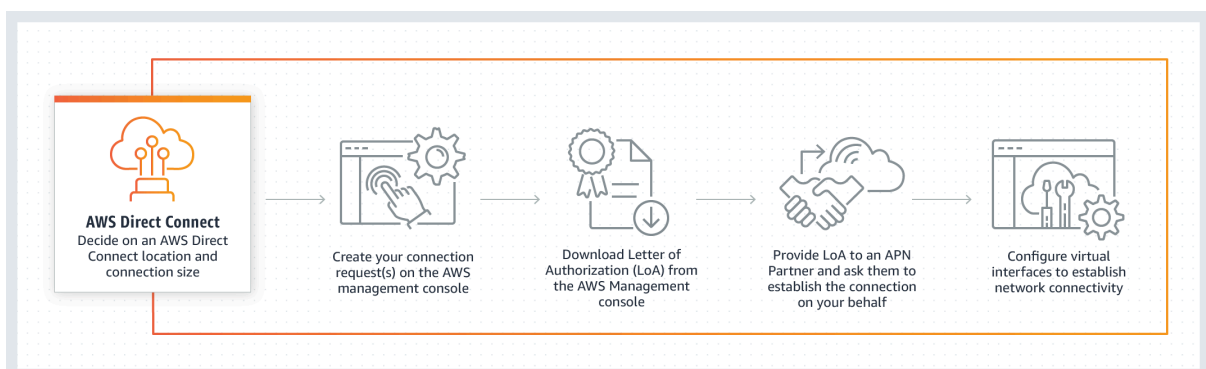
Overall explanation

Correct option:

AWS Direct Connect

AWS Direct Connect is a cloud service solution that makes it easy to establish a dedicated network connection from your premises to AWS. Using AWS Direct Connect, you can establish private connectivity between AWS and your data center, office, or colocation environment, which in many cases can reduce your network costs, increase bandwidth throughput, and provide a more consistent network experience than Internet-based connections.

How AWS Direct Connect works:



via - <https://aws.amazon.com/directconnect/>

Incorrect options:

AWS Site-to-Site VPN - By default, instances that you launch into an Amazon VPC can't communicate with your own (remote) network. You can enable access to your remote network from your VPC by creating an AWS Site-to-Site VPN (Site-to-Site VPN) connection and configuring routing to pass traffic through the connection. It uses the public internet and is therefore not suited for this use case.

VPC Endpoints - A VPC endpoint enables you to privately connect your VPC to supported AWS services and VPC endpoint services powered by AWS PrivateLink without requiring an internet gateway, NAT device, VPN connection, or AWS Direct Connect connection. It does not connect your on-premises data centers and AWS Cloud.

VPC peering connection - A VPC peering connection is a networking connection between two virtual private clouds (VPCs) that enables you to route traffic between them using private IPv4 addresses or IPv6 addresses. It is used to connect virtual private clouds (VPCs) together, and not on-premises data centers and AWS Cloud.

Reference:

<https://aws.amazon.com/directconnect/>

Domain

Security and Compliance

Question 31 Incorrect

Which AWS service allows you to quickly and easily add user sign-up, sign-in, and access control to web and mobile applications?

Your answer is incorrect

AWS Identity and Access Management (AWS IAM)

Correct answer

Amazon Cognito

AWS IAM Identity Center

AWS Organizations

Overall explanation

Correct option:

Amazon Cognito

Amazon Cognito lets you add user sign-up, sign-in, and access control to your web and mobile apps quickly and easily. With Amazon Cognito, you also have the option to authenticate users through social identity providers such as Facebook, Twitter, or Amazon, with SAML identity solutions, or by using your own identity system.

Incorrect options:

AWS Identity and Access Management (AWS IAM) - AWS Identity and Access Management (AWS IAM) enables you to securely control access to AWS services and resources for your users. Using IAM, you can create and manage AWS users and groups, and use permissions to allow and deny their access to AWS resources. It does not provide user sign-up, sign-in, and access control to web and mobile applications.

AWS IAM Identity Center - AWS IAM Identity Center is the successor to AWS Single Sign-On. It is built on top of AWS Identity and Access Management (IAM) to simplify access management to multiple AWS accounts, AWS applications, and other SAML-enabled cloud applications. In IAM Identity Center, you create, or connect, your workforce users for use across AWS. You can choose to manage access just to your AWS accounts, just to your cloud applications, or to both.

AWS Organizations - AWS Organizations offers policy-based management for multiple AWS accounts. With AWS Organizations, you can create groups of accounts, automate account creation, and apply and manage policies for those groups. Organizations enable you to centrally manage policies across multiple accounts, without requiring custom scripts and manual processes. It does not provide user sign-up, sign-in, and access control to web and mobile applications.

Reference:

<https://aws.amazon.com/cognito/>

Domain

Technology

Question 32Correct

A company needs to use a secure online data transfer tool/service that can automate the ongoing transfers from on-premises systems into AWS while providing support for incremental data backups.

Which AWS tool/service is an optimal fit for this requirement?

AWS Storage Gateway

AWS Snowball Edge

AWS Snowball

Your answer is correct

AWS DataSync

Overall explanation

Correct option:

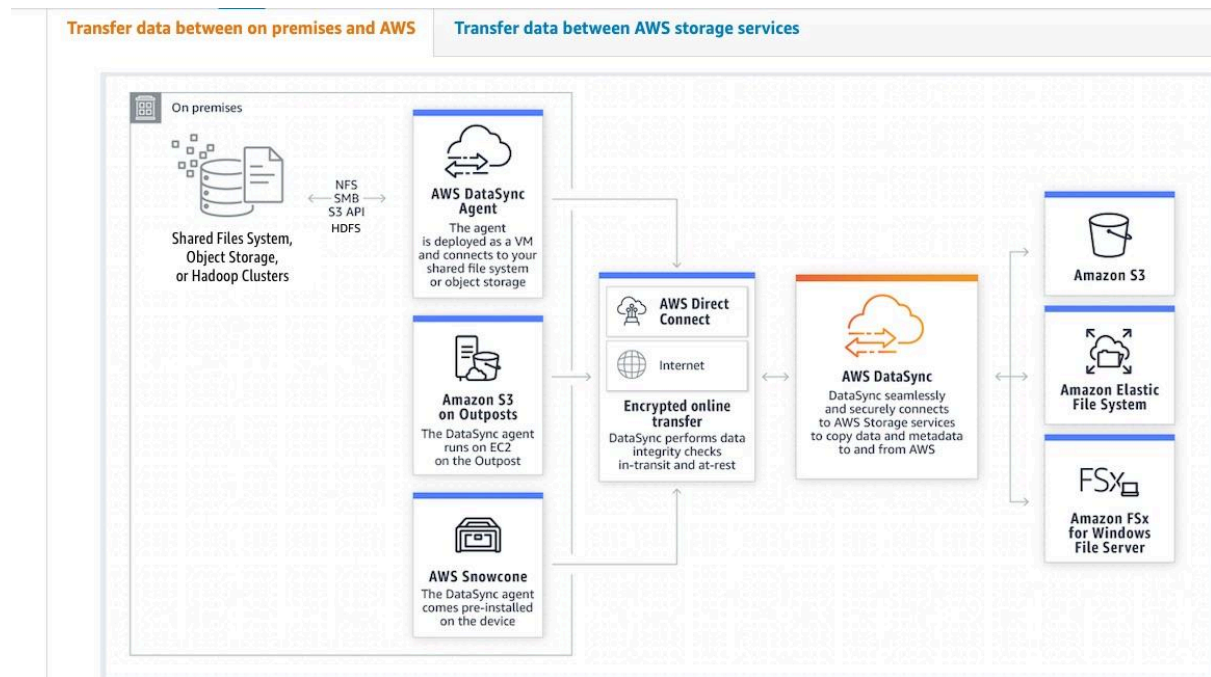
AWS DataSync

AWS DataSync is a secure online data transfer service that simplifies, automates, and accelerates copying terabytes of data to and from AWS storage services. Easily migrate or replicate large data sets without having to build custom solutions or oversee repetitive tasks. DataSync can copy data between Network File System (NFS) shares, or Server Message Block (SMB) shares, self-managed object storage, AWS Snowcone, Amazon Simple Storage Service (Amazon S3) buckets, Amazon Elastic File System (Amazon EFS) file systems, and Amazon FSx for Windows File Server file systems.

You can use AWS DataSync for ongoing transfers from on-premises systems into or out of AWS for processing. DataSync can help speed up your critical hybrid cloud storage workflows in industries that need to move active files into AWS quickly. This includes machine learning in life sciences, video production in media and entertainment, and big data analytics in financial services. AWS DataSync provides timely delivery to ensure dependent processes are not delayed. You can specify exclude filters, include filters, or both, to determine which files, folders, or objects get transferred each time your task runs.

AWS DataSync employs an AWS-designed transfer protocol—decoupled from the storage protocol—to accelerate data movement. The protocol performs optimizations on how, when, and what data is sent over the network. Network optimizations performed by DataSync include incremental transfers, in-line compression, and sparse file detection, as well as in-line data validation and encryption.

Data Transfer between on-premises and AWS using AWS DataSync:



via - <https://aws.amazon.com/datasync/>

Incorrect options:

AWS Storage Gateway - AWS Storage Gateway is a set of hybrid cloud services that give you on-premises access to virtually unlimited cloud storage. Customers use AWS Storage Gateway to integrate AWS Cloud storage with existing on-site workloads so they can simplify storage management and reduce costs for key hybrid cloud storage use cases. These include moving backups to the cloud, using on-premises file shares backed by cloud storage, and providing low latency access to data in AWS for on-premises applications.

AWS Snowball - AWS Snowball is a physical data transport device used for large-scale offline data transfers. It is ideal when you need to migrate terabytes or petabytes of data and lack high-speed internet. However, it is not an online or ongoing transfer solution, nor does it support automated or incremental backups.

AWS Snowball Edge - AWS Snowball Edge is an enhanced version of AWS Snowball that supports local compute capabilities in addition to offline data transfers. It is suitable for edge computing and disconnected environments, not for ongoing online data transfers with incremental sync.

References:

<https://aws.amazon.com/datasync/>

<https://aws.amazon.com/datasync/features/>

<https://aws.amazon.com/snowball/>

<https://docs.aws.amazon.com/snowball/latest/developer-guide/whatisedge.html>

Domain

Technology

Question 33Correct

Which of the following are the advantages of using the AWS Cloud? (Select TWO)

AWS is responsible for security in the cloud

Your selection is correct

Increase speed and agility

Your selection is correct

Stop guessing about capacity

Limited scaling

Trade operational expense for capital expense

Overall explanation

Correct options:

Increase speed and agility

Stop guessing about capacity

Exam Alert:

Please check out the following six advantages of Cloud Computing. You would certainly be asked questions on the advantages of Cloud Computing compared to a traditional on-premises setup:

Six Advantages of Cloud Computing

PDF  | RSS

- **Trade capital expense for variable expense** – Instead of having to invest heavily in data centers and servers before you know how you're going to use them, you can pay only when you consume computing resources, and pay only for how much you consume.
- **Benefit from massive economies of scale** – By using cloud computing, you can achieve a lower variable cost than you can get on your own. Because usage from hundreds of thousands of customers is aggregated in the cloud, providers such as AWS can achieve higher economies of scale, which translates into lower pay as-you-go prices.
- **Stop guessing capacity** – Eliminate guessing on your infrastructure capacity needs. When you make a capacity decision prior to deploying an application, you often end up either sitting on expensive idle resources or dealing with limited capacity. With cloud computing, these problems go away. You can access as much or as little capacity as you need, and scale up and down as required with only a few minutes' notice.
- **Increase speed and agility** – In a cloud computing environment, new IT resources are only a click away, which means that you reduce the time to make those resources available to your developers from weeks to just minutes. This results in a dramatic increase in agility for the organization, since the cost and time it takes to experiment and develop is significantly lower.
- **Stop spending money running and maintaining data centers** – Focus on projects that differentiate your business, not the infrastructure. Cloud computing lets you focus on your own customers, rather than on the heavy lifting of racking, stacking, and powering servers.
- **Go global in minutes** – Easily deploy your application in multiple regions around the world with just a few clicks. This means you can provide lower latency and a better experience for your customers at minimal cost.

via -

<https://docs.aws.amazon.com/whitepapers/latest/aws-overview/six-advantages-of-cloud-computing.html>

Incorrect options:

Limited scaling - Scaling is not limited in the cloud. You can access as much or as little capacity as you need, and scale up and down as required with only a few minutes' notice.

AWS is responsible for security in the cloud - AWS is responsible for the security OF the cloud, which means AWS is responsible for protecting the infrastructure that runs all the services offered in the AWS Cloud.

Trade operational expense for capital expense - In the cloud, you trade capital expense (CAPEX) for the operational expense (OPEX). Instead of having to invest heavily in data centers and servers before you know how you're going to use them, you can pay only when you consume computing resources, and pay only for how much you consume.

Reference:

<https://docs.aws.amazon.com/whitepapers/latest/aws-overview/six-advantages-of-cloud-computing.html>

Domain

Cloud Concepts

Question 34Correct

A growing start-up has trouble identifying and protecting sensitive data at scale.
Which AWS fully managed service can assist with this task?

AWS Artifact

AWS Secrets Manager

Your answer is correct

Amazon Macie

AWS Key Management Service (AWS KMS)

Overall explanation

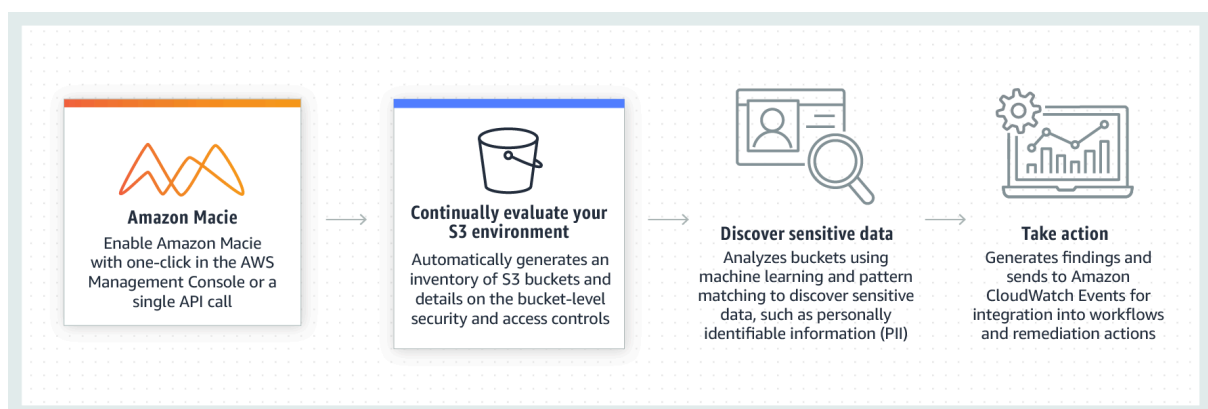
Correct option:

Amazon Macie

Amazon Macie is a fully managed data security and data privacy service that uses Machine Learning and pattern matching to discover and protect your sensitive data in AWS.

Amazon Macie uses Machine Learning and pattern matching to cost-efficiently discover sensitive data at scale. Amazon Macie automatically detects a large and growing list of sensitive data types, including personally identifiable information (PII) such as names, addresses, and credit card numbers. It also gives you constant visibility of the data security and data privacy of your data stored in Amazon S3.

How Amazon Macie works:



via - <https://aws.amazon.com/macie/>

Incorrect options:

AWS Artifact - AWS Artifact is your go-to, central resource for compliance-related information that matters to you. It provides on-demand access to AWS' security and

compliance reports and selects online agreements. It is not used to discover and protect sensitive data in AWS.

AWS Secrets Manager - AWS Secrets Manager helps you protect secrets needed to access your applications, services, and IT resources. The service enables you to easily rotate, manage, and retrieve database credentials, API keys, and other secrets throughout their lifecycle. It is not used to discover and protect sensitive data in AWS.

AWS Key Management Service (AWS KMS) - AWS Key Management Service (AWS KMS) makes it easy for you to create and manage keys and control the use of encryption across a wide range of AWS services and in your applications. It is not used to discover and protect sensitive data in AWS.

Reference:

<https://aws.amazon.com/macie/>

Domain

Security and Compliance

Question 35Correct

According to the AWS Well-Architected Framework, which of the following action is recommended in the Security pillar?

Use AWS CloudFormation to automate security best practices

Use AWS Cost Explorer to view and track your usage in detail

Your answer is correct

Use AWS Key Management Service (AWS KMS) to encrypt data

Use Amazon CloudWatch to measure overall efficiency

Overall explanation

Correct option:

Use AWS Key Management Service (AWS KMS) to encrypt data

The Security pillar includes the ability to protect information, systems, and assets while delivering business value through risk assessments and mitigation strategies.

Encrypting data is part of the design principle "Protect data in transit and at rest":
Classify your data into sensitivity levels and use mechanisms, such as encryption, tokenization, and access control where appropriate.

AWS Key Management Service (AWS KMS) makes it easy for you to create and control keys used for encryption. It is a key service of the Security pillar.

The AWS Well-Architected Framework helps you understand the pros and cons of the decisions you make while building systems on AWS. By using the Framework you will learn architectural best practices for designing and operating reliable, secure, efficient, and cost-effective systems in the cloud. It provides a way for you to consistently measure your architectures against best practices and identify areas for improvement.

The AWS Well-Architected Framework is based on six pillars – Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, and Sustainability.

Overview of the six pillars of the AWS Well-Architected Framework:

AWS Well-Architected and the Six Pillars

Framework Overview

The AWS Well-Architected Framework describes key concepts, design principles, and architectural best practices for designing and running workloads in the cloud. By answering a few foundational questions, learn how well your architecture aligns with cloud best practices and gain guidance for making improvements.

[HTML](#) | [Kindle](#) | [Labs](#)



Operational Excellence Pillar The operational excellence pillar focuses on running and monitoring systems, and continually improving processes and procedures. Key topics include automating changes, responding to events, and defining standards to manage daily operations. HTML Kindle Labs	Security Pillar The security pillar focuses on protecting information and systems. Key topics include confidentiality and integrity of data, managing user permissions, and establishing controls to detect security events. HTML Kindle Labs	Reliability Pillar The reliability pillar focuses on workloads performing their intended functions and how to recover quickly from failure to meet demands. Key topics include distributed system design, recovery planning, and adapting to changing requirements. HTML Kindle Labs
Performance Efficiency Pillar The performance efficiency pillar focuses on structured and streamlined allocation of IT and computing resources. Key topics include selecting resource types and sizes optimized for workload requirements, monitoring performance, and maintaining efficiency as business needs evolve.	Cost Optimization Pillar The cost optimization pillar focuses on avoiding unnecessary costs. Key topics include understanding spending over time and controlling fund allocation, selecting resources of the right type and quantity, and scaling to meet business needs without overspending.	Sustainability Pillar The sustainability pillar focuses on minimizing the environmental impacts of running cloud workloads. Key topics include a shared responsibility model for sustainability, understanding impact, and maximizing utilization to minimize required resources and reduce downstream impacts.

via - <https://aws.amazon.com/architecture/well-architected/>

Incorrect options:

Use AWS Cost Explorer to view and track your usage in detail - AWS Cost Explorer has an easy-to-use interface that lets you visualize, understand, and manage your AWS costs and usage over time. Using Cost Explorer to view and track your usage in detail relates more to the Cost Optimization pillar.

Use Amazon CloudWatch to measure overall efficiency - Amazon CloudWatch is a monitoring and management service built for developers, system operators, site reliability engineers (SRE), and IT managers. Using Amazon CloudWatch to measure overall efficiency relates more to the Reliability pillar.

Use AWS CloudFormation to automate security best practices - AWS

CloudFormation provides a common language for you to model and provision AWS and third-party application resources in your cloud environment. It is not used to automate security best practices. If you want to automate security best practices, you should use Amazon Inspector.

Reference:

<https://aws.amazon.com/architecture/well-architected/>

Domain

Cloud Concepts

Question 36Correct

A production company with predictable usage would like to reduce the cost of its Amazon Elastic Compute Cloud (Amazon EC2) instances by using reserved instances (RI). Which of the following length terms are available for Amazon EC2 reserved instances (RI)? (Select Two)

6 months

Your selection is correct

1 year

5 years

Your selection is correct

3 years

2 years

Overall explanation

Correct options:

1 year

3 years

Reserved Instances (RI) provide you with a significant discount (up to 75%) compared to On-Demand instance pricing. Besides, when Reserved Instances (RI) are assigned to a specific Availability Zone (AZ), they provide a capacity reservation,

giving you additional confidence in your ability to launch instances when you need them.

Standard and Convertible reserved instances can be purchased for a 1-year or 3-year term.

EC2 Pricing Options Overview:

On-Demand

With On-Demand instances, you pay for compute capacity by the hour or the second depending on which instances you run. No longer-term commitments or upfront payments are needed. You can increase or decrease your compute capacity depending on the demands of your application and only pay the specified per hourly rates for the instance you use.

On-Demand instances are recommended for:

- Users that prefer the low cost and flexibility of Amazon EC2 without any up-front payment or long-term commitment
- Applications with short-term, spiky, or unpredictable workloads that cannot be interrupted
- Applications being developed or tested on Amazon EC2 for the first time

[See On-Demand pricing »](#)

Spot instances

Amazon EC2 Spot instances allow you to request spare Amazon EC2 computing capacity for up to 90% off the On-Demand price. [Learn More.](#)

Spot instances are recommended for:

- Applications that have flexible start and end times
- Applications that are only feasible at very low compute prices
- Users with urgent computing needs for large amounts of additional capacity

[See Spot pricing »](#)

Savings Plans

Savings Plans are a flexible pricing model that offer low prices on EC2 and Fargate usage, in exchange for a commitment to a consistent amount of usage (measured in \$/hour) for a 1 or 3 year term.

Dedicated Hosts

A Dedicated Host is a physical EC2 server dedicated for your use. Dedicated Hosts can help you reduce costs by allowing you to use your existing server-bound software licenses, including Windows Server, SQL Server, and SUSE Linux Enterprise Server (subject to your license terms), and can also help you meet compliance requirements. [Learn more.](#)

- Can be purchased On-Demand (hourly).
- Can be purchased as a Reservation for up to 70% off the On-Demand price.

[See Dedicated pricing »](#)

Reserved Instances

Reserved Instances provide you with a significant discount (up to 75%) compared to On-Demand instance pricing. In addition, when Reserved Instances are assigned to a specific Availability Zone, they provide a capacity reservation, giving you additional confidence in your ability to launch instances when you need them.

For applications that have steady state or predictable usage, Reserved Instances can provide significant savings compared to using On-Demand instances. See [How to Purchase Reserved Instances](#) for more information.

Reserved Instances are recommended for:

- Applications with steady state usage
- Applications that may require reserved capacity
- Customers that can commit to using EC2 over a 1 or 3 year term to reduce their total computing costs

via - <https://aws.amazon.com/ec2/pricing/>

Incorrect options:

6 months - It is not possible to reserve instances for 6 months.

5 years - It is not possible to reserve instances for 5 years.

2 years - It is not possible to reserve instances for 2 years.

Reference:

<https://aws.amazon.com/ec2/pricing/>

Domain

Technology

Question 37Correct

An organization would like to copy data across different Availability Zones (AZs) using Amazon EBS snapshots. Where are Amazon EBS snapshots stored in the AWS Cloud?

Amazon Elastic Compute Cloud (Amazon EC2)

Your answer is correct

Amazon Simple Storage Service (Amazon S3)

Amazon Elastic File System (Amazon EFS)

Amazon Relational Database Service (Amazon RDS)

Overall explanation

Correct option:

Amazon Simple Storage Service (Amazon S3)

You can create a point-in-time snapshot of an Amazon EBS Elastic Volume and use it as a baseline for new volumes or data backup. If you make periodic snapshots of a volume, the snapshots are incremental—the new snapshot saves only the blocks that have changed since your last snapshot.

You can back up the data on your Amazon EBS Elastic Volumes to Amazon Simple Storage Service (Amazon S3) by taking point-in-time snapshots.

Incorrect options:

Amazon Elastic Compute Cloud (Amazon EC2) - Amazon Elastic Compute Cloud (Amazon EC2) is a web service that provides secure, resizable compute capacity in the cloud. It is designed to make web-scale computing easier for developers. Amazon EBS snapshots cannot be stored on Amazon EC2.

Amazon Relational Database Service (Amazon RDS) - Amazon Relational Database Service (Amazon RDS) makes it easy to set up, operate, and scale a relational database in the cloud. Amazon EBS snapshots cannot be stored on Amazon RDS.

Amazon Elastic File System (Amazon EFS) - Amazon Elastic File System (Amazon EFS) provides a simple, scalable, elastic file system for Linux-based workloads for use with AWS Cloud services and on-premises resources. Amazon EBS snapshots cannot be stored on Amazon EFS.

Reference:

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/EBSSnapshots.html>

Domain

Technology

Question 38Correct

A company based in Sydney hosts its application on an Amazon Elastic Compute Cloud (Amazon EC2) instance in ap-southeast-2. They would like to deploy the same Amazon EC2 instances in eu-south-1. Which of the following AWS entities can address this use case?

Your answer is correct

Amazon Machine Image (AMI)

Amazon EBS Elastic Volume snapshots

AWS Lambda

Elastic Load Balancing (ELB)

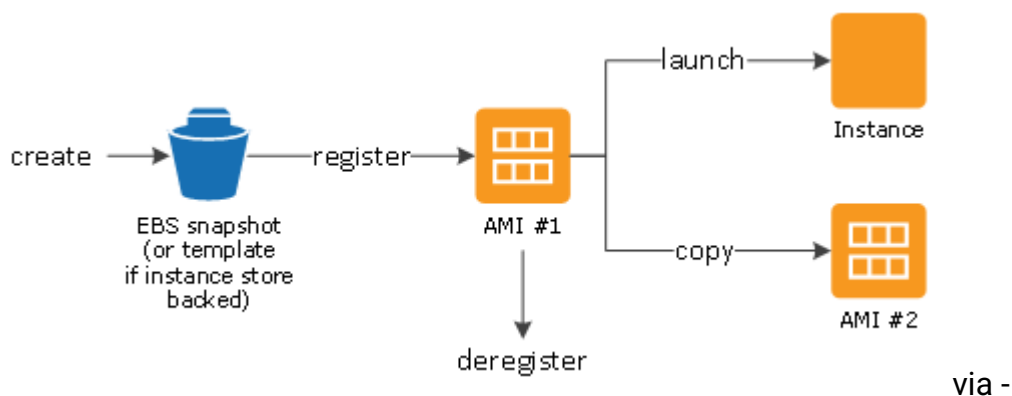
Overall explanation

Correct option:

Amazon Machine Image (AMI)

An Amazon Machine Image (AMI) provides the information required to launch an instance. You must specify an Amazon Machine Image (AMI) when you launch an instance. You can launch multiple instances from a single Amazon Machine Image (AMI) when you need multiple instances with the same configuration.

How to use an Amazon Machine Image (AMI):



via -

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/AMIs.html>

Incorrect options:

Elastic Load Balancing (ELB) - Elastic Load Balancing (ELB) automatically distributes incoming application traffic across multiple targets, such as Amazon EC2 instances,

containers, IP addresses, and Lambda functions. It can handle the varying load of your application traffic in a single Availability Zone (AZ) or across multiple Availability Zones (AZs). It cannot be used to deploy the same EC2 instances across different Availability Zones (AZs).

AWS Lambda - AWS Lambda lets you run code without provisioning or managing servers. It cannot be used to deploy the same EC2 instances across different Availability Zones (AZs).

Amazon EBS Elastic Volume snapshots - An Amazon EBS snapshot is a point-in-time copy of your Amazon EBS volume. EBS snapshots are one of the components of an AMI, but EBS snapshots alone cannot be used to deploy the same EC2 instances across different Availability Zones (AZs).

Reference:

<https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/AMIs.html>

Domain

Technology

Question 39Incorrect

A Cloud Practitioner would like to get operational insights of its resources to quickly identify any issues that might impact applications using those resources. Which AWS service can help with this task?

Correct answer

AWS Systems Manager

Amazon Inspector

Your answer is incorrect

AWS Health Dashboard - Your Account Health

AWS Trusted Advisor

Overall explanation

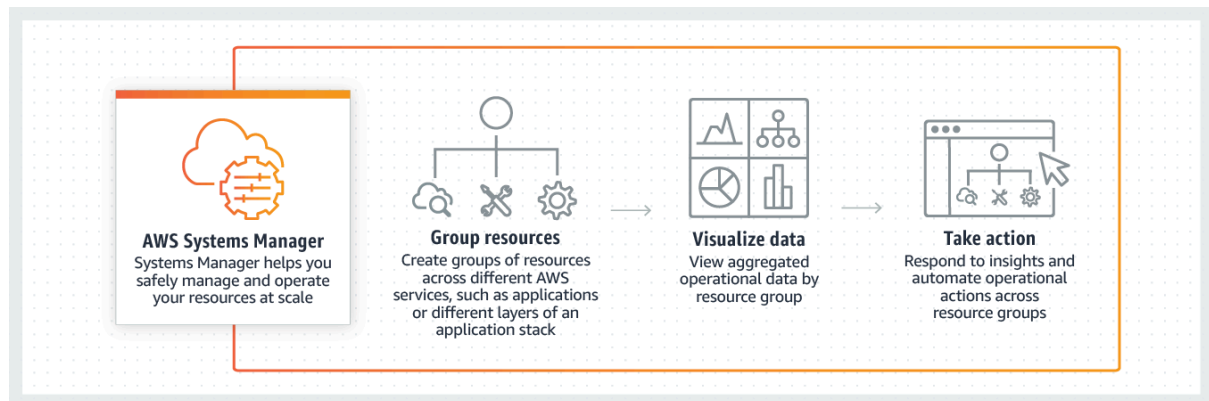
Correct option:

AWS Systems Manager

AWS Systems Manager allows you to centralize operational data from multiple AWS services and automate tasks across your AWS resources. You can create logical groups of resources such as applications, different layers of an application stack, or production versus development environments.

With AWS Systems Manager, you can select a resource group and view its recent API activity, resource configuration changes, related notifications, operational alerts, software inventory, and patch compliance status. You can also take action on each resource group depending on your operational needs. AWS Systems Manager provides a central place to view and manage your AWS resources, so you can have complete visibility and control over your operations.

How AWS Systems Manager works:



via - <https://aws.amazon.com/systems-manager/>

Incorrect options:

Amazon Inspector - Amazon Inspector is an automated security assessment service that helps improve the security and compliance of applications deployed on AWS. It is not used to get operational insights of AWS resources.

AWS Health Dashboard - Your Account Health - AWS Health Dashboard - Your Account Health provides alerts and remediation guidance when AWS is experiencing events that may impact you. It is not used to get operational insights of AWS resources.

AWS Trusted Advisor - AWS Trusted Advisor is an online resource to help you reduce cost, increase performance, and improve security by optimizing your AWS environment. AWS Trusted Advisor provides real-time guidance to help you provision your resources following AWS best practices. It is not used to get operational insights of AWS resources.

Reference:

<https://aws.amazon.com/systems-manager/>

Domain

Security and Compliance

Question 40Incorrect

Which of the following are the best practices when using AWS Organizations?

(Select TWO)

Your selection is correct

Restrict account privileges using Service Control Policies (SCP)

Correct selection

Create AWS accounts per department

Disable AWS CloudTrail on several accounts

Your selection is incorrect

Do not use AWS Organizations to automate AWS account creation

Never use tags for billing

Overall explanation

Correct options:

Create AWS accounts per department

Restrict account privileges using Service Control Policies (SCP)

AWS Organizations helps you centrally govern your environment as you grow and scale your workloads on AWS. Whether you are a growing startup or a large enterprise, AWS Organizations help you to centrally manage billing; control access, compliance, and security; and share resources across your AWS accounts.

Using AWS Organizations, you can automate account creation, create groups of accounts to reflect your business needs, and apply policies for these groups for governance. You can also simplify billing by setting up a single payment method for all of your AWS accounts. Through integrations with other AWS services, you can use AWS Organizations to define central configurations and resource sharing across accounts in your organization. AWS Organizations is available to all AWS customers at no additional charge.

You should create accounts per department based on regulatory restrictions (using Service Control Policies (SCP)) for better resource isolation, and to have separate per-account service limits.

AWS Organizations allows you to restrict what services and actions are allowed in your accounts. You can use the Service Control Policies (SCP) to apply permission guardrails on AWS Identity and Access Management (IAM) users and roles.

Incorrect options:

Never use tags for billing - You should use tags standards to categorize AWS resources for billing purposes.

Disable AWS CloudTrail on several accounts - You should enable AWS CloudTrail to monitor activity on all accounts for governance, compliance, risk, and auditing purposes.

Do not use AWS Organizations to automate AWS account creation - AWS Organizations helps you simplify IT operations by automating AWS account creation and management. The AWS Organizations APIs enable you to create new accounts programmatically and to add new accounts to a group. The policies attached to the group are automatically applied to the new account.

Reference:

<https://aws.amazon.com/organizations/>

Domain

Billing and Pricing

Question 41Correct

A brand-new startup would like to remove its need to manage the underlying infrastructure and focus on the deployment and management of its applications.

Which type of cloud computing does this refer to?

Software as a Service (SaaS)

Infrastructure as a Service (IaaS)

Your answer is correct

Platform as a Service (PaaS)

On-premises

Overall explanation

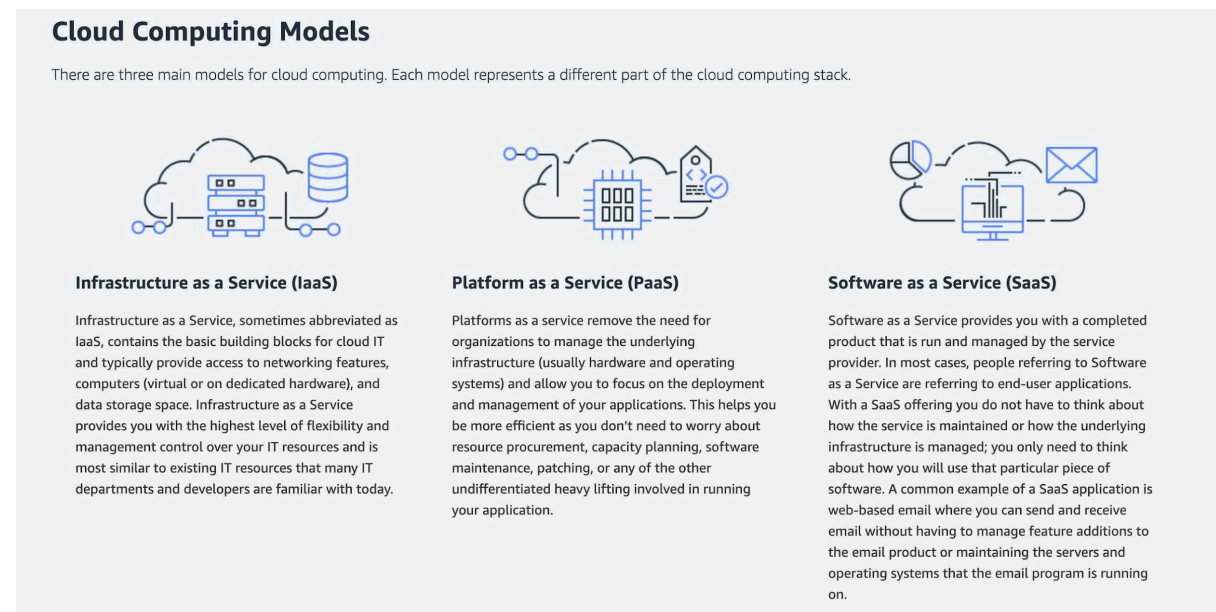
Correct option:

Platform as a Service (PaaS)

Cloud Computing can be broadly divided into three types - Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS).

Platform as a Service (PaaS) removes the need to manage underlying infrastructure (usually hardware and operating systems) and allows you to focus on the deployment and management of your applications. You don't need to worry about resource procurement, capacity planning, software maintenance, patching, or any of the other undifferentiated heavy lifting involved in running your application.

Please review this overview of the types of cloud computing:



via - <https://aws.amazon.com/types-of-cloud-computing/>

Incorrect options:

Infrastructure as a Service (IaaS) - Infrastructure as a Service (IaaS) contains the basic building blocks for cloud IT. It typically provides access to networking features, computers (virtual or on dedicated hardware), and data storage space. Infrastructure as a Service (IaaS) gives the highest level of flexibility and management control over IT resources.

Software as a Service (SaaS) - Software as a Service (SaaS) provides you with a complete product that is run and managed by the service provider. With a Software as a Service (SaaS) offering, you don't have to think about how the service is maintained or how the underlying infrastructure is managed. You only need to think about how you will use that particular software. Amazon Rekognition is an example of a SaaS service.

On-premises - When an enterprise opts for on-premises, it needs to create, upgrade, and scale the on-premise IT infrastructure by investing in sophisticated hardware, compatible software, and robust services. Also, the business needs to deploy dedicated IT staff to upkeep, scale, and manage the on-premise infrastructure continuously.

Reference:

<https://aws.amazon.com/types-of-cloud-computing/>

Domain

Cloud Concepts

Question 42Incorrect

Which of the following services are provided by Amazon Route 53? (Select Two)

Correct selection

Health checks and monitoring

Load balancing

Your selection is incorrect

IP routing

Your selection is correct

Domain registration

Transfer acceleration

Overall explanation

Correct options:

Domain registration

Health checks and monitoring

Amazon Route 53 is a highly available and scalable cloud Domain Name System (DNS) web service. It is designed to give developers and businesses an extremely reliable and cost-effective way to route end users to Internet applications by translating names like `www.example.com` into the numeric IP addresses like `192.0.2.1` that computers use to connect to each other.

Amazon Route 53 offers domain name registration services, where you can search for and register available domain names or transfer in existing domain names to be managed by Route 53.

Amazon Route 53 can monitor the health and performance of your application as well as your web servers and other resources.

Incorrect options:

IP routing - Despite its name, Amazon Route 53 does not offer IP routing. However, it can route traffic based on multiple criteria, such as endpoint health, geographic location, and latency, using routing policies.

Load balancing - It is a feature of Elastic Load Balancing (ELB) and not Amazon Route 53. Elastic Load Balancing (ELB) automatically distributes incoming application traffic across multiple targets, such as Amazon EC2 instances, containers, IP addresses, and Lambda functions. It can handle the varying load of your application traffic in a single Availability Zone (AZ) or across multiple Availability Zones (AZs).

Transfer acceleration - Transfer acceleration is a feature of Amazon's simple storage service (Amazon S3). Amazon S3 Transfer Acceleration (Amazon S3TA) can speed up content transfers to and from Amazon S3 by as much as 50-500% for long-distance transfer of larger objects.

Reference:

<https://aws.amazon.com/route53/>

Domain

Technology

Question 43Correct

A corporation would like to simplify access management to multiple AWS accounts as well as facilitate AWS Single Sign-On (AWS SSO) access to its AWS accounts. As a Cloud Practitioner, which AWS service would you use for this task?

AWS Cognito

AWS Identity and Access Management (AWS IAM)

Your answer is correct

AWS IAM Identity Center

AWS Command Line Interface (CLI)

Overall explanation

Correct option:

AWS IAM Identity Center

AWS IAM Identity Center is the successor to AWS Single Sign-On (AWS SSO). It is built on top of AWS Identity and Access Management (IAM) to simplify access

management to multiple AWS accounts, AWS applications, and other SAML-enabled cloud applications. In IAM Identity Center, you create or connect, your workforce users for use across AWS. You can choose to manage access just to your AWS accounts, just to your cloud applications, or to both.

You can create users directly in IAM Identity Center, or you can bring them from your existing workforce directory. With IAM Identity Center, you get a unified administration experience to define, customize, and assign fine-grained access. Your workforce users get a user portal to access their assigned AWS accounts or cloud applications.

You can use IAM Identity Center to quickly and easily assign and manage your employees' access to multiple AWS accounts, SAML-enabled cloud applications (such as Salesforce, Microsoft 365, and Box), and custom-built in-house applications, all from a central place.

How AWS IAM Identity Center works:



via - <https://aws.amazon.com/iam/identity-center/>

Incorrect options:

AWS Cognito - Amazon Cognito lets you add user sign-up, sign-in, and access control to your web and mobile apps quickly and easily. With Amazon Cognito, you also have the option to authenticate users through social identity providers such as Facebook, Twitter, or Amazon, with SAML identity solutions, or by using your own identity system. It is an identity management solution for customers/developers building B2C or B2B apps for their customers.

AWS Identity and Access Management (AWS IAM) - AWS Identity and Access Management (AWS IAM) enables you to securely control access to AWS services and resources for your users. Using AWS IAM, you can create and manage AWS users and groups, and use permissions to allow and deny their access to AWS resources. It is not used to log in but to manage users and roles.

AWS Command Line Interface (CLI) - The AWS Command Line Interface (CLI) is a unified tool to manage your AWS services. With just one tool to download and configure, you can control multiple AWS services from the command line and automate them through scripts. It is not a central user portal.

Reference:

<https://aws.amazon.com/iam/identity-center/>

Domain

Technology

Question 44Correct

Which of the following criteria are used to calculate the charge for Amazon EBS Volumes? (Select Two)

Data transfer IN

Data type

The Amazon EC2 instance type the Amazon EBS Elastic volume is attached to

Your selection is correct

Provisioned IOPS

Your selection is correct

Volume type

Overall explanation

Correct options:

Provisioned IOPS

Volume type

Amazon Elastic Block Store (Amazon EBS) provides persistent block storage volumes for use with Amazon EC2 instances in the AWS Cloud. Each Amazon EBS volume is automatically replicated within its Availability Zone (AZ) to protect you from component failure, offering high availability and durability. Amazon EBS volumes offer the consistent and low-latency performance needed to run your

workloads. With Amazon EBS, you can scale your usage up or down within minutes—all while paying a low price for only what you provision.

The fundamental charges for EBS volumes are the volume type (based on performance), the storage volume in GB per month provisioned, the number of IOPS provisioned per month, the storage consumed by snapshots, and outbound data transfer.

Incorrect options:

Data transfer IN - Data transfer-in is always free, including for Amazon EBS Elastic Volumes.

The Amazon EC2 instance type the Amazon EBS Elastic volume is attached to - The Amazon EC2 instance type the Amazon EBS volume is attached to does not influence the EBS volume pricing.

Data type - The type of data stored on EBS volumes does not influence the price.

Reference:

<https://aws.amazon.com/ebs/pricing/>

Domain

Billing and Pricing

Question 45Correct

Which of the following statements is CORRECT regarding the scope of an Amazon Virtual Private Cloud (VPC)?

A VPC spans all Availability Zones (AZs) in all AWS regions

Amazon VPC spans all subnets in all AWS regions

A VPC spans all AWS regions within an Availability Zone (AZ)

Your answer is correct

A VPC spans all Availability Zones (AZs) within an AWS region

Overall explanation

Correct option:

A VPC spans all Availability Zones (AZs) within an AWS region

Amazon Virtual Private Cloud (Amazon VPC) lets you provision a logically isolated section of the AWS Cloud where you can launch AWS resources in a virtual network that you define. You have complete control over your virtual networking environment,

including selection of your own IP address range, creation of subnets, and configuration of route tables and network gateways.

An Amazon VPC spans all Availability Zones (AZs) within a region.

Incorrect options:

Amazon VPC spans all subnets in all AWS regions - A VPC is located within an AWS region.

A VPC spans all Availability Zones (AZs) in all AWS regions - A VPC is located within an AWS region.

A VPC spans all AWS regions within an Availability Zone (AZ) - AWS has the concept of a Region, which is a physical location around the world where AWS clusters data centers. Each AWS Region consists of multiple (two or more), isolated, and physically separate Availability Zone (AZs) within a geographic area. An Availability Zone (AZ) is one or more discrete data centers with redundant power, networking, and connectivity in an AWS Region. Therefore, regions cannot be within an Availability Zone. Moreover, a VPC is located within a region.

AWS Regions and Availability Zones (AZs) Overview:

Regions

AWS has the concept of a Region, which is a physical location around the world where we cluster data centers. We call each group of logical data centers an Availability Zone. Each AWS Region consists of multiple, isolated, and physically separate AZ's within a geographic area. Unlike other cloud providers, who often define a region as a single data center, the multiple AZ design of every AWS Region offers advantages for customers. Each AZ has independent power, cooling, and physical security and is connected via redundant, ultra-low-latency networks. AWS customers focused on high availability can design their applications to run in multiple AZ's to achieve even greater fault-tolerance. AWS infrastructure Regions meet the highest levels of security, compliance, and data protection.

AWS provides a more extensive global footprint than any other cloud provider, and to support its global footprint and ensure customers are served across the world, AWS opens new Regions rapidly. AWS maintains multiple geographic Regions, including Regions in North America, South America, Europe, China, Asia Pacific, South Africa, and the Middle East.

Availability Zones

An Availability Zone (AZ) is one or more discrete data centers with redundant power, networking, and connectivity in an AWS Region. AZ's give customers the ability to operate production applications and databases that are more highly available, fault tolerant, and scalable than would be possible from a single data center. All AZ's in an AWS Region are interconnected with high-bandwidth, low-latency networking, over fully redundant, dedicated metro fiber providing high-throughput, low-latency networking between AZ's. All traffic between AZ's is encrypted. The network performance is sufficient to accomplish synchronous replication between AZ's. AZ's make partitioning applications for high availability easy. If an application is partitioned across AZ's, companies are better isolated and protected from issues such as power outages, lightning strikes, tornadoes, earthquakes, and more. AZ's are physically separated by a meaningful distance, many kilometers, from any other AZ, although all are within 100 km (60 miles) of each other.

via - https://aws.amazon.com/about-aws/global-infrastructure/regions_az/

Reference:

<https://aws.amazon.com/vpc/>

Domain

Technology

Question 46Correct

An e-commerce company would like to build a chatbot for its customer service using Natural Language Understanding (NLU). As a Cloud Practitioner, which AWS service would you use?

Your answer is correct

Amazon Lex

Amazon Rekognition

Amazon Comprehend

Amazon SageMaker

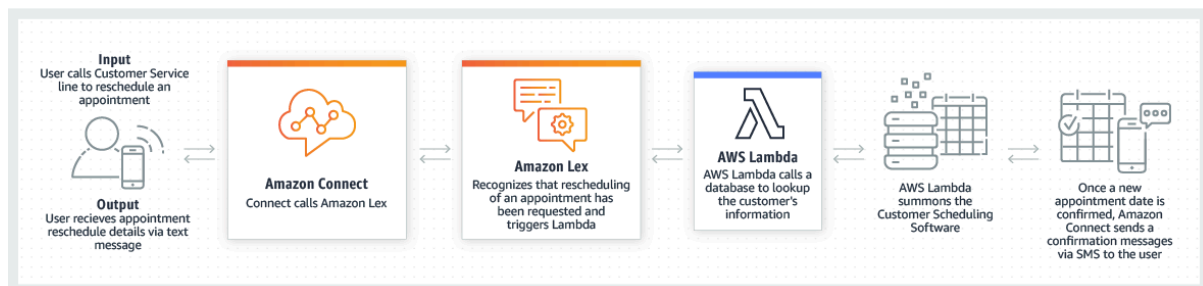
Overall explanation

Correct option:

Amazon Lex

Amazon Lex is a service for building conversational interfaces using voice and text. Powered by the same conversational engine as Amazon Alexa, Amazon Lex provides high-quality speech recognition and language understanding capabilities, enabling the addition of sophisticated, natural language 'chatbots' to new and existing applications.

Amazon Lex Use Cases:



via - <https://aws.amazon.com/lex/>

Incorrect options:

Amazon Rekognition - With Amazon Rekognition, you can identify objects, people, text, scenes, and activities in images and videos and also detect any inappropriate content. Amazon Rekognition also provides highly accurate facial analysis and facial

search capabilities that you can use to detect, analyze, and compare faces for a wide variety of user verification, people counting, and public safety use cases.

Amazon SageMaker - Amazon SageMaker is a fully-managed platform that enables developers and data scientists to quickly and easily build, train, and deploy machine learning models at any scale. Amazon SageMaker removes all the barriers that typically slow down developers who want to use machine learning.

Amazon Comprehend - Amazon Comprehend is a natural language processing (NLP) service that uses machine learning to find meaning and insights in text. Natural Language Processing (NLP) is a way for computers to analyze, understand, and derive meaning from textual information in a smart and useful way. By utilizing Natural Language Processing (NLP), you can extract important phrases, sentiment, syntax, key entities such as brand, date, location, person, etc., and the language of the text.

Reference:

<https://aws.amazon.com/lex/>

Domain

Technology

Question 47Correct

Which AWS service can be used to view the most comprehensive billing details for the past month?

AWS Pricing Calculator

Your answer is correct

AWS Cost & Usage Report (AWS CUR)

AWS Cost Explorer

AWS Budgets

Overall explanation

Correct option:

AWS Cost & Usage Report (AWS CUR)

The AWS Cost & Usage Report (AWS CUR) contains the most comprehensive set of cost and usage data available. You can use Cost and Usage Reports to publish your AWS billing reports to an Amazon Simple Storage Service (Amazon S3) bucket that you own. You can receive reports that break down your costs by the hour or month, by product or product resource, or by tags that you define yourself.

AWS Cost & Usage Report (AWS CUR) Overview:

What are AWS Cost and Usage Reports?

[PDF](#) | [RSS](#)

The AWS Cost and Usage Reports (AWS CUR) contains the most comprehensive set of cost and usage data available. You can use Cost and Usage Reports to publish your AWS billing reports to an Amazon Simple Storage Service (Amazon S3) bucket that you own. You can receive reports that break down your costs by the hour or month, by product or product resource, or by tags that you define yourself. AWS updates the report in your bucket once a day in comma-separated value (CSV) format. You can view the reports using spreadsheet software such as Microsoft Excel or Apache OpenOffice Calc, or access them from an application using the Amazon S3 API.

AWS Cost and Usage Reports tracks your AWS usage and provides estimated charges associated with your account. Each report contains line items for each unique combination of AWS products, usage type, and operation that you use in your AWS account. You can customize the AWS Cost and Usage Reports to aggregate the information either by the hour or by the day.

AWS Cost and Usage Reports can do the following:

- Deliver report files to your Amazon S3 bucket
- Update the report up to three times a day
- Create, retrieve, and delete your reports using the AWS CUR API Reference

via - <https://docs.aws.amazon.com/cur/latest/userguide/what-is-cur.html>

Incorrect options:

AWS Budgets - AWS Budgets gives the ability to set custom budgets that alert you when your costs or usage exceed (or are forecasted to exceed) your budgeted amount. You can also use AWS Budgets to set reservation utilization or coverage targets and receive alerts when your utilization drops below the threshold you define. Budgets can be created at the monthly, quarterly, or yearly level, and you can customize the start and end dates. You can further refine your budget to track costs associated with multiple dimensions, such as AWS service, linked account, tag, and others. AWS Budgets cannot provide billing details for the past month.

AWS Cost Explorer - AWS Cost Explorer has an easy-to-use interface that lets you visualize, understand, and manage your AWS costs and usage over time. AWS Cost Explorer includes a default report that helps you visualize the costs and usage associated with your top five cost-accruing AWS services and gives you a detailed breakdown of all services in the table view. The reports let you adjust the time range to view historical data going back up to twelve months to gain an understanding of your cost trends. AWS Cost Explorer cannot provide granular billing details for the past month.

AWS Pricing Calculator - AWS Pricing Calculator lets you explore AWS services and create an estimate for the cost of your use cases on AWS. You can model your solutions before building them, explore the price points and calculations behind your estimate, and find the available instance types and contract terms that meet your needs. This enables you to make informed decisions about using AWS. You can plan your AWS costs and usage or price out by setting up a new set of instances and services. AWS Pricing Calculator cannot provide billing details for the past month.

Exam Alert:

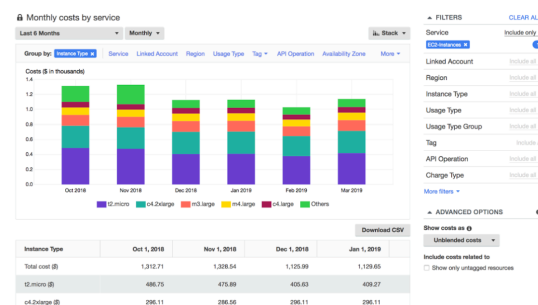
Please review the differences between "AWS Cost & Usage Report (AWS CUR)" and "AWS Cost Explorer". Think of "AWS Cost & Usage Report (AWS CUR)" as a cost management tool providing the most detailed cost and usage data for your AWS account. It can provide reports that break down your costs by the hour into your Amazon Simple Storage Service (Amazon S3) bucket. On the other hand, "AWS Cost Explorer" is more of a high-level cost management tool that helps you visualize the costs and usage associated with your AWS account.

"AWS Cost Explorer" vs "AWS Cost & Usage Report (AWS CUR)":

Monthly Costs by AWS Service

AWS Cost Explorer includes a default report that helps you visualize the costs and usage associated with your top five cost-accruing AWS services, and gives you a detailed breakdown on all services in the table view. The reports let you adjust the time range to view historical data going back up to twelve months to gain an understanding of your cost trends.

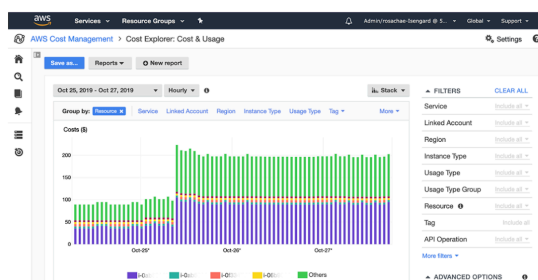
[Launch the Monthly Costs by AWS Service report »](#)



Hourly and Resource Level Granularity

AWS Cost Explorer helps you visualize, understand, and manage your AWS costs and usage over a daily or monthly granularity. The solution also lets you dive deeper using granular filtering and grouping dimensions such as Usage Type and Tags. You can also access your data with further granularity by enabling hourly and resource level granularity.

[Get started using Hourly and Resource Level Granularity »](#)



via - <https://aws.amazon.com/aws-cost-management/aws-cost-explorer/>

How does the AWS Cost & Usage Report work?

AWS delivers the AWS Cost & Usage Report (in CSV format) to whichever Amazon Simple Storage Service (S3) bucket you specify, and updates the reports at least once per day. You can download any of the reports using the Amazon S3 console, or you can retrieve the reports programmatically using the Amazon S3 APIs.

You can configure your Cost & Usage Reports to integrate with Amazon Athena. Once Amazon Athena integration has been enabled for your Cost & Usage Report, your data will be delivered in compressed Apache Parquet files to an Amazon S3 bucket of your choice. Your AWS Cost & Usage Report can also be ingested directly into Amazon Redshift or uploaded to Amazon QuickSight.

	M	N	O	P	Q	R	S	T
1	InstanceUsage	InstanceUsage	Usage	Usage	Usage	Usage	Usage	Usage
2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2
3	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2
4	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2
5	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2
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30	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2
31	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2	AmazonEC2

via -

<https://aws.amazon.com/aws-cost-management/aws-cost-and-usage-reporting/>

References:

<https://docs.aws.amazon.com/cur/latest/userguide/what-is-cur.html>

<https://aws.amazon.com/aws-cost-management/aws-cost-explorer/>

<https://aws.amazon.com/aws-cost-management/aws-cost-and-usage-reporting/>

Domain

Billing and Pricing

Question 48Correct

The IT infrastructure at a university is deployed on AWS Cloud and it's experiencing a read-intensive workload. As a Cloud Practitioner, which AWS service would you use to take the load off databases?

Your answer is correct

Amazon ElastiCache

Amazon Relational Database Service (Amazon RDS)

AWS Glue

Amazon EMR

Overall explanation

Correct option:

Amazon ElastiCache

Amazon ElastiCache is a web service that makes it easy to deploy, operate, and scale an in-memory cache in the cloud. The service improves the performance of web applications by allowing you to retrieve information from fast, managed, in-memory caches, instead of relying entirely on slower disk-based databases.

If Amazon EC2 instances are intensively reading data from a database, ElastiCache can cache some values to take the load off the database.

How Amazon ElastiCache works:



via - <https://aws.amazon.com/elasticache/>)

Incorrect options:

Amazon Relational Database Service (Amazon RDS) - Amazon Relational Database Service (Amazon RDS) makes it easy to set up, operate, and scale a relational database in the cloud. It provides cost-efficient and resizable capacity while automating time-consuming administration tasks such as hardware provisioning, database setup, patching, and backups. It frees you to focus on your applications so you can give them the fast performance, high availability, security, and compatibility they need. It cannot be used to take the load off databases. However, Amazon ElastiCache is often used with Amazon RDS to take the load off RDS.

AWS Glue - AWS Glue is a fully managed extract, transform, and load (ETL) service that makes it easy for customers to prepare and load their data for analytics. AWS Glue job is meant to be used for batch ETL data processing. It cannot be used to take the load off the databases.

Amazon EMR - Amazon EMR provides a managed Hadoop framework that makes it easy, fast, and cost-effective to process vast amounts of data across dynamically scalable Amazon EC2 instances. It cannot be used to take the load off the databases.

Reference:

<https://aws.amazon.com/elasticache/>

Domain

Technology

Question 49Correct

Which of the following AWS services can be used to generate, use, and manage encryption keys on the AWS Cloud?

Amazon Inspector

AWS GuardDuty

Your answer is correct

AWS CloudHSM

AWS Secrets Manager

Overall explanation

Correct option:

AWS CloudHSM

The AWS CloudHSM service helps you meet corporate, contractual, and regulatory compliance requirements for data security by using dedicated Hardware Security Module (HSM) instances within the AWS cloud.

AWS CloudHSM allows you to securely generate, store, and manage cryptographic keys used for data encryption in a way that keys are accessible only to you.

How AWS CloudHSM works:



via - <https://aws.amazon.com/cloudhsm/>

Incorrect options:

Amazon Inspector - Amazon Inspector is an automated security assessment service that helps improve the security and compliance of applications deployed on AWS. Amazon Inspector automatically assesses applications for exposure, vulnerabilities, and deviations from best practices. It cannot be used to generate, use, and manage encryption keys.

AWS GuardDuty - Amazon GuardDuty is a threat detection service that continuously monitors for malicious or unauthorized behavior to help you protect your AWS accounts and workloads. It cannot be used to generate, use, and manage encryption keys.

AWS Secrets Manager - AWS Secrets Manager helps you protect secrets needed to access your applications, services, and IT resources. The service enables you to

easily rotate, manage, and retrieve database credentials, API keys, and other secrets throughout their lifecycle. It is integrated with AWS CloudHSM to generate, use, and manage encryption keys.

Reference:

<https://aws.amazon.com/cloudhsm/>

Domain

Security and Compliance

Question 50Correct

An engineering team would like to cost-effectively run hundreds of thousands of batch computing workloads on AWS. As a Cloud Practitioner, which AWS service would you use for this task?

AWS Fargate

Amazon Lightsail

Your answer is correct

AWS Batch

AWS Lambda

Overall explanation

Correct option:

AWS Batch

AWS Batch enables developers, scientists, and engineers to easily and efficiently run hundreds of thousands of batch computing jobs on AWS.

You can use AWS Batch to plan, schedule, and execute your batch computing workloads across the full range of AWS compute services. AWS Batch dynamically provisions the optimal quantity and type of compute resources (for example - memory optimized instance or CPU) based on the volume and specific resource requirements of the batch jobs submitted. AWS Batch provisions compute resources and optimize the job distribution based on the volume and resource requirements of the submitted batch jobs.

Please review the common use cases for AWS Batch:

Use cases

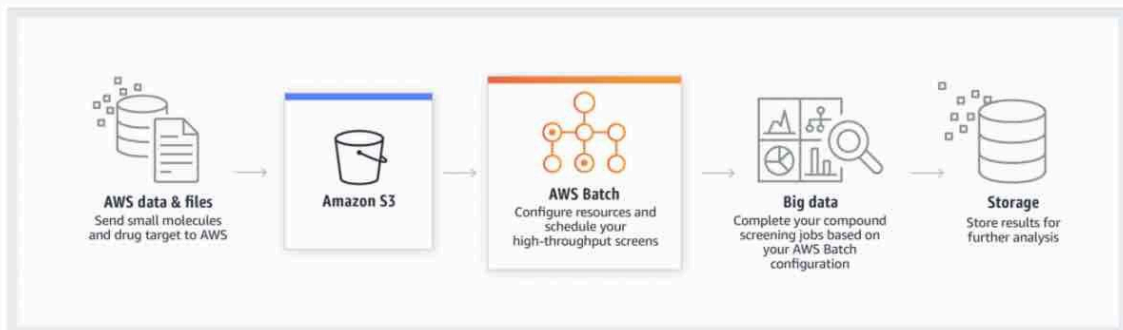
Financial services: Post-trade analytics

Automate the analysis of the day's transaction costs, execution reporting, and market performance.



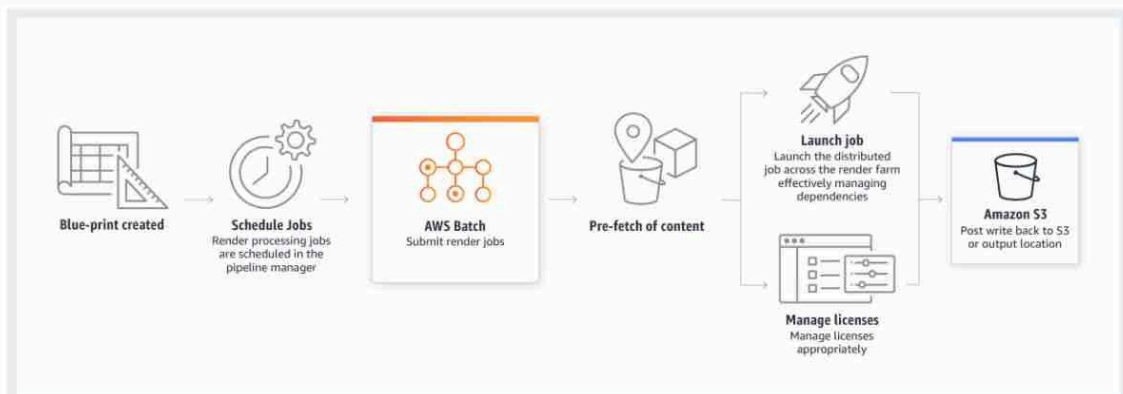
Life sciences: Drug screening for biopharma

Rapidly search libraries of small molecules for drug discovery.



Digital media: Visual effects rendering

Automate content rendering workloads and reduce the need for human intervention due to execution dependencies or resource scheduling.



via - <https://aws.amazon.com/batch/>

Incorrect options:

AWS Lambda - AWS Lambda lets you run code without provisioning or managing servers. It can be used to run batch jobs but has a time limit and limited runtimes. It is usually used for smaller batch jobs.

Amazon Lightsail - Amazon Lightsail is designed to be the easiest way to launch and manage a virtual private server (VPS) with AWS. Amazon Lightsail plans include everything you need to jumpstart your project – a virtual machine, SSD- based storage, data transfer, Domain Name System (DNS) management, and a static IP address – for a low, predictable price. It is not used to run batch jobs.

AWS Fargate - AWS Fargate is a compute engine for Amazon Elastic Container Service (Amazon ECS) that allows you to run containers without having to manage servers or clusters. You can run batch jobs on AWS Fargate, but it is more expensive than AWS Batch.

Reference:

<https://aws.amazon.com/batch/>

Domain

Technology

Question 51 Correct

A company would like to audit requests made to an Amazon Simple Storage Service (Amazon S3) bucket. As a Cloud Practitioner, which Amazon Simple Storage Service (Amazon S3) feature would you recommend addressing this use-case?

Your answer is correct

Amazon Simple Storage Service (Amazon S3) Access Logs

S3 Versioning

Amazon S3 Bucket Policies

S3 cross-region replication (S3 CRR)

Overall explanation

Correct option:

Amazon Simple Storage Service (Amazon S3) Access Logs

Server access logging provides detailed records for the requests that are made to an Amazon S3 bucket. Server access logs are useful for many applications. For example, access log information can be useful in security and access audits.

It can also help you learn about your customer base and understand your Amazon S3 bill.

Incorrect options:

S3 cross-region replication (S3 CRR) - S3 cross-region replication (S3 CRR) enables automatic, asynchronous copying of objects across Amazon S3 buckets.

Cross-Region replication (CRR) is used to copy objects across Amazon S3 buckets in different AWS Regions. It does not help with auditing requests made to your bucket.

Amazon S3 Bucket Policies - Amazon S3 Bucket policy is a resource-based AWS Identity and Access Management (IAM) policy. You add a bucket policy to a bucket to grant other AWS accounts or IAM users access permissions for the bucket and the objects in it. Object permissions apply only to the objects that the bucket owner creates. It does not help with auditing requests made to your bucket.

S3 Versioning - Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. It does not help with auditing requests made to your bucket.

Reference:

<https://docs.aws.amazon.com/AmazonS3/latest/dev/ServerLogs.html>

Domain

Security and Compliance

Question 52Correct

Which AWS serverless service allows you to prepare data for analytics?

Amazon Redshift

Your answer is correct

AWS Glue

Amazon Athena

Amazon EMR

Overall explanation

Correct option:

AWS Glue

AWS Glue is a fully managed extract, transform, and load (ETL) service that makes it easy for customers to prepare and load their data for analytics. AWS Glue job is meant to be used for batch ETL data processing.

How AWS Glue works:



via - <https://aws.amazon.com/glue/>

Incorrect options:

Amazon Athena - Amazon Athena is an interactive query service that makes it easy to analyze data in Amazon Simple Storage Service (Amazon S3) using standard SQL. Athena is serverless, so there is no infrastructure to manage, and you pay only for the queries that you run. Amazon Athena is used for analytics and not to prepare data for analytics.

Amazon Redshift - Amazon Redshift is a fast and scalable data warehouse that makes it simple and cost-effective to analyze all your data across your data warehouse and data lake. Redshift is used for analytics and not to prepare data for analytics.

Amazon EMR - Amazon EMR provides a managed Hadoop framework that makes it easy, fast, and cost-effective to process vast amounts of data across dynamically scalable Amazon EC2 instances. EMR is used for analytics and not to prepare data for analytics.

Reference:

<https://aws.amazon.com/glue/>

Domain

Technology

Question 53Correct

A company needs to keep sensitive data in its own data center due to compliance but would still like to deploy resources using AWS. Which Cloud deployment model does this refer to?

Private Cloud

On-premises

Public Cloud

Your answer is correct

Hybrid Cloud

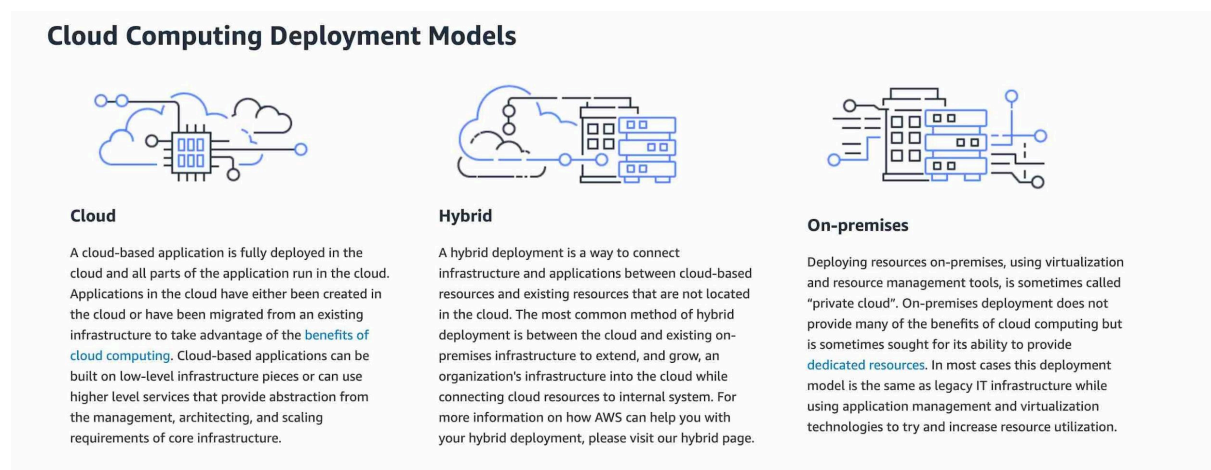
Overall explanation

Correct option:

Hybrid Cloud

A hybrid deployment is a way to connect infrastructure and applications between cloud-based resources and existing resources that are not located in the cloud. The most common method of hybrid deployment is between the cloud and existing on-premises infrastructure to extend, and grow, an organization's infrastructure into the cloud while connecting cloud resources to the internal system.

Overview of Cloud Computing Deployment Models:



via - <https://aws.amazon.com/types-of-cloud-computing/>

Incorrect options:

Public Cloud - A public cloud-based application is fully deployed in the cloud and all parts of the application run in the cloud. Applications in the cloud have either been created in the cloud or have been migrated from an existing infrastructure to take advantage of the benefits of cloud computing.

Private Cloud - Unlike a Public cloud, a Private cloud enables businesses to avail IT services that are provisioned and customized according to their precise needs. The business can further avail the IT services securely and reliably over a private IT infrastructure.

On-premises - This is not a cloud deployment model. When an enterprise opts for on-premises, it needs to create, upgrade, and scale the on-premise IT infrastructure by investing in sophisticated hardware, compatible software, and robust services. Also, the business needs to deploy dedicated IT staff to upkeep, scale, and manage the on-premise infrastructure continuously.

Reference:

<https://docs.aws.amazon.com/whitepapers/latest/public-sector-cloud-transformation/selecting-the-right-cloud-for-workloads-differences-between-public-private-and-hybrid.html>

Domain

Cloud Concepts

Question 54Incorrect

A company using a hybrid cloud would like to store secondary backup copies of the on-premises data. Which Amazon S3 Storage Class would you use for a cost-optimal yet rapid access solution?

Amazon S3 Standard

Correct answer

Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA)

Your answer is incorrect

Amazon S3 Standard-Infrequent Access (S3 Standard-IA)

Amazon S3 Glacier Deep Archive

Overall explanation

Correct option:

Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA)

Amazon S3 One Zone-Infrequent Access (S3 One Zone-IA) is for data that is accessed less frequently but requires rapid access when needed. Unlike other S3 Storage Classes which store data in a minimum of three Availability Zones (AZs), S3 One Zone-IA stores data in a single Availability Zone (AZ) and costs 20% less than S3 Standard-IA. S3 One Zone-IA is ideal for customers who want a lower-cost option for infrequently accessed data but do not require the availability and resilience of S3 Standard or S3 Standard-IA. It's a good choice for storing secondary backup copies of on-premises data or easily re-creatable data. You can also use it as cost-effective storage for data that is replicated from another AWS Region using S3 cross-region replication (S3 CRR).

Exam Alert:

Please review this detailed comparison of S3 Storage Classes as you can expect a few questions on this aspect of S3:

Performance across the S3 Storage Classes

	S3 Standard	S3 Intelligent-Tiering*	S3 Standard-IA	S3 One Zone-IA†	S3 Glacier Instant Retrieval	S3 Glacier Flexible Retrieval	S3 Glacier Deep Archive
Designed for durability	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)	99.999999999% (11 9's)
Designed for availability	99.99%	99.9%	99.9%	99.5%	99.9%	99.99%	99.99%
Availability SLA	99.9%	99%	99%	99%	99%	99.9%	99.9%
Availability Zones	≥3	≥3	≥3	1	≥3	≥3	≥3
Minimum capacity charge per object	N/A	N/A	128 KB	128 KB	128 KB	40 KB	40 KB
Minimum storage duration charge	N/A	N/A	30 days	30 days	90 days	90 days	180 days
Retrieval charge	N/A	N/A	per GB retrieved	per GB retrieved	per GB retrieved	per GB retrieved	per GB retrieved
First byte latency	milliseconds	milliseconds	milliseconds	milliseconds	milliseconds	minutes or hours	hours
Storage type	Object	Object	Object	Object	Object	Object	Object
Lifecycle transitions	Yes	Yes	Yes	Yes	Yes	Yes	Yes

via - <https://aws.amazon.com/s3/storage-classes/>

Incorrect options:

Amazon S3 Glacier Deep Archive - Amazon S3 Glacier Deep Archive storage class is designed to provide durable and secure long-term storage for large amounts of data at a price that is competitive with off-premises tape archival services. Data is stored across 3 or more AWS Availability Zones(AZs) and can be retrieved in 12 hours or less. You no longer need to deal with expensive and finicky tape drives, arrange for off-premises storage, or worry about migrating data to newer generations of media.

Amazon S3 Standard - Amazon S3 Standard offers high durability, availability, and performance object storage for frequently accessed data. Because it delivers low latency and high throughput, Amazon S3 Standard is appropriate for a wide variety of use cases, including cloud applications, dynamic websites, content distribution, mobile and gaming applications, and big data analytics.

Amazon S3 Standard-Infrequent Access (S3 Standard-IA) - Amazon S3 Standard-Infrequent Access (S3 Standard-IA) is for data that is accessed less frequently but requires rapid access when needed. S3 Standard-IA offers the high durability, high throughput, and low latency of S3 Standard, with a low per GB storage price and per GB retrieval fee. This combination of low cost and high performance makes S3 Standard-IA ideal for long-term storage, backups, and as a data store for disaster recovery files. It can be used for backups, but it is more expensive than S3

One Zone - Infrequent Access. Hence, S3 One Zone - Infrequent Access is a better option for secondary backup copies.

Reference:

<https://aws.amazon.com/s3/storage-classes/>

Domain

Technology

Question 55Correct

A company is planning to implement Chaos Engineering to expose any blind spots that can disrupt the resiliency of the application.

Which AWS service will help implement this requirement with the least effort?

Amazon GuardDuty

AWS Trusted Advisor

Amazon Inspector

Your answer is correct

AWS Fault Injection Simulator (AWS FIS)

Overall explanation

Correct option:

AWS Fault Injection Simulator (AWS FIS)

AWS Fault Injection Simulator (AWS FIS) is a fully managed service for running fault injection experiments on AWS that makes it easier to improve an application's performance, observability, and resiliency. Fault injection experiments are used in chaos engineering, which is the practice of stressing an application in testing or production environments by creating disruptive events, such as a sudden increase in CPU or memory consumption, observing how the system responds, and implementing improvements. Fault injection experiment helps teams create the real-world conditions needed to uncover the hidden bugs, and monitor blind spots, and performance bottlenecks that are difficult to find in distributed systems.

AWS Fault Injection Simulator (AWS FIS) simplifies the process of setting up and running controlled fault injection experiments across a range of AWS services so teams can build confidence in their application behavior. With AWS Fault Injection Simulator (AWS FIS), teams can quickly set up experiments using pre-built templates that generate the desired disruptions. AWS Fault Injection Simulator (AWS FIS) provides the controls and guardrails that teams need to run experiments in

production, such as automatically rolling back or stopping the experiment if specific conditions are met. With a few clicks in the console, teams can run complex scenarios with common distributed system failures happening in parallel or building sequentially over time, enabling them to create the real-world conditions necessary to find hidden weaknesses.

Incorrect options:

Amazon Inspector - Amazon Inspector is an automated security assessment service that helps improve the security and compliance of applications deployed on AWS. Amazon Inspector automatically assesses applications for exposure, vulnerabilities, and deviations from best practices. After performing an assessment, Amazon Inspector produces a detailed list of security findings prioritized by level of severity.

AWS Trusted Advisor - AWS Trusted Advisors provides recommendations that help you follow AWS best practices. AWS Trusted Advisor evaluates your account by using checks. These checks identify ways to optimize your AWS infrastructure, improve security and performance, reduce costs, and monitor service quotas. You can then follow the check recommendations to optimize your services and resources.

Amazon GuardDuty - Amazon GuardDuty is a threat detection service that continuously monitors your AWS accounts and workloads for malicious activity and delivers detailed security findings for visibility and remediation.

Reference:

<https://aws.amazon.com/fis/features/>

Domain

Technology

Question 56Correct

Which of the following AWS Support plans is the MOST cost-effective when getting enhanced technical support by Cloud Support Engineers?

AWS Enterprise Support

AWS Basic Support

AWS Developer Support

Your answer is correct

AWS Business Support

Overall explanation

Correct option:

AWS Business Support

AWS recommends AWS Business Support if you have production workloads on AWS and want 24x7 phone, email, and chat access to technical support and architectural guidance in the context of your specific use cases. You get full access to AWS Trusted Advisor Best Practice Checks. It is also the cheapest support plan to provide enhanced technical support by Cloud Support Engineers.

Exam Alert:

Please review the differences between the AWS Developer Support, AWS Business Support, and AWS Enterprise Support plans as you can expect at least a couple of questions on the exam:

	Developer	Business	Enterprise On-Ramp	Enterprise
	<i>Recommended if you are experimenting or testing in AWS.</i>	<i>Minimum recommended tier if you have production workloads in AWS</i>	<i>Recommended if you have production and/or business critical workloads in AWS.</i>	<i>Recommended if you have business and/or mission critical workloads in AWS.</i>
AWS Trusted Advisor Best Practice Checks	Service Quota and basic Security checks	Full set of checks	Full set of checks	Full set of checks
AWS Trusted Advisor Priority				Prioritized recommendations curated by your AWS account team
Enhanced Technical Support	Business hours** web access to Cloud Support Associates Unlimited cases with 1 primary contact Prioritized responses on AWS re:Post	24/7 phone, web, and chat access to Cloud Support Engineers Unlimited cases and unlimited contacts (IAM supported) Prioritized responses on AWS re:Post Access to AWS Support App in Slack	24/7 phone, web, and chat access to Cloud Support Engineers Unlimited cases and unlimited contacts (IAM supported) Prioritized responses on AWS re:Post Access to AWS Support App in Slack	24/7 phone, web, and chat access to Cloud Support Engineers Unlimited cases and unlimited contacts (IAM supported) Prioritized responses on AWS re:Post Access to AWS Support App in Slack
Case Severity / Response Times*	General guidance: < 24 hours** System impaired: < 12 hours**	General guidance: < 24 hours System impaired: < 12 hours Production system impaired: < 4 hours Production system down: < 1 hour	General guidance: < 24 hours System impaired: < 12 hours Production system impaired: < 4 hours Production system down: < 1 hour Business-critical system down: < 30 minutes	General guidance: < 24 hours System impaired: < 12 hours Production system impaired: < 4 hours Production system down: < 1 hour Business/Mission-critical system down: < 15 minutes
Architectural Guidance	General	Contextual to your use-cases	Consultative review and guidance based on your applications (one-per-year)	Consultative review and guidance based on your applications
Programmatic Case Management		AWS Support API	AWS Support API	AWS Support API

	Developer	Business	Enterprise On-Ramp	Enterprise
	<i>Recommended if you are experimenting or testing in AWS.</i>	<i>Minimum recommended tier if you have production workloads in AWS</i>	<i>Recommended if you have production and/or business critical workloads in AWS.</i>	<i>Recommended if you have business and/or mission critical workloads in AWS.</i>
Third-Party Software Support		Interoperability and configuration guidance and troubleshooting	Interoperability and configuration guidance and troubleshooting	Interoperability and configuration guidance and troubleshooting
Proactive Programs and Self Service	Access to Support Automation Workflows with prefixes AWSsupport	Access to Infrastructure Event Management for additional fee Access to Support Automation Workflows with prefixes AWSsupport and AWSPremiumSupport	Infrastructure Event Management (one-per-year) Access to Support Automation Workflows with prefixes AWSsupport and AWSPremiumSupport	Access to Infrastructure Event Management Access to proactive reviews, workshops, and deep dives Access to Support Automation Workflows with prefixes AWSsupport and AWSPremiumSupport
AWS Incident Detection and Response				Access to AWS Incident Detection and Response for an additional fee. AWS Incident Detection and Response is an add-on to Enterprise Support that offers 24x7 proactive monitoring and incident management for selected workloads. AWS Incident Detection and Response leverages the proven operational, enhanced monitoring, and incident management capabilities used internally by AWS teams and externally by AWS Managed Services (AMS).
AWS Managed Services		Access to AWS Managed Services (AMS) for an additional fee. AMS augments your existing teams with cloud advanced operations skills and capacity. Includes baseline operations, a designated Cloud Service Delivery Manager (CSDM), Cloud Architect (CA), and access to the AMS security team.	Access to AWS Managed Services (AMS) for an additional fee. AMS augments your existing teams with cloud advanced operations skills and capacity. Includes baseline operations, a designated Cloud Service Delivery Manager (CSDM), Cloud Architect (CA), and access to the AMS security team.	Access to AWS Managed Services (AMS) for an additional fee. AMS augments your existing teams with cloud operations skills and capacity. It includes baseline operations, a designated Cloud Service Delivery Manager (CSDM), Cloud Architect (CA), and access to the AMS security team. AWS Incident Detection and Response is available at no additional charge in eligible regions for AWS Managed Services direct customers with AWS Enterprise Support.

	Developer	Business	Enterprise On-Ramp	Enterprise
	<i>Recommended if you are experimenting or testing in AWS.</i>	<i>Minimum recommended tier if you have production workloads in AWS</i>	<i>Recommended if you have production and/or business critical workloads in AWS.</i>	<i>Recommended if you have business and/or mission critical workloads in AWS.</i>
Technical Account Management			A pool of Technical Account Managers to provide proactive guidance, and coordinate access to programs and AWS experts	Designated Technical Account Manager (TAM) to proactively monitor your environment and assist with optimization and coordinate access to programs and AWS experts
Training				Access to online self-paced labs
Account Assistance			Concierge Support Team	Concierge Support Team
Pricing	Greater of \$29 / month*** - or - 3% of monthly AWS usage See pricing detail and example.	Greater of \$100 / month*** - or - 10% of monthly AWS usage for the first \$0–\$10K 7% of monthly AWS usage from \$10K–\$80K 5% of monthly AWS usage from \$80K–\$250K 3% of monthly AWS usage over \$250K See pricing detail and example.	Greater of \$5,500 - or - 10% of monthly AWS usage See pricing detail and example.	Greater of \$15,000 - or - 10% of monthly AWS usage for the first \$0–\$150K 7% of monthly AWS usage from \$150K–\$500K 5% of monthly AWS usage from \$500K–\$1M 3% of monthly AWS usage over \$1M See pricing detail and example.
*Additional services for additional fee		*Access to AWS Managed Services (AMS) for an additional fee	*Access to AWS Managed Services (AMS) for an additional fee	* Access to AWS Incident Detection and Response for an additional fee. *Access to AWS Managed Services (AMS) for an additional fee

via - <https://aws.amazon.com/premiumsupport/plans/>

Incorrect options:

AWS Developer Support - AWS recommends AWS Developer Support if you are testing or doing early development on AWS and want the ability to get technical support during business hours as well as general architectural guidance as you build and test. It provides enhanced technical support by Cloud Support Associates.

AWS Basic Support - The AWS Basic Support plan is included for all AWS customers. It does not provide enhanced technical support.

AWS Enterprise Support - AWS Enterprise Support provides customers with concierge-like service where the main focus is helping the customer achieve their outcomes and find success in the cloud. With Enterprise Support, you get 24x7 technical support from high-quality engineers, tools, and technology to automatically manage the health of your environment, consultative architectural guidance delivered in the context of your applications and use cases, and a designated Technical Account Manager (TAM) to coordinate access to proactive/preventative

programs and AWS subject matter experts. It provides enhanced technical support by Cloud Support Engineers but is more expensive than the Business support plan.

References:

<https://aws.amazon.com/premiumsupport/plans/>

<https://aws.amazon.com/premiumsupport/plans/business/>

Domain

Billing and Pricing

Question 57

A data science team would like to build Machine Learning models for its projects. Which AWS service can it use?

Amazon Comprehend

Your answer is correct

Amazon SageMaker

Amazon Polly

Amazon Connect

Overall explanation

Correct option:

Amazon SageMaker

Amazon SageMaker is a fully-managed platform that enables developers and data scientists to quickly and easily build, train, and deploy machine learning models at any scale. Amazon SageMaker removes all the barriers that typically slow down developers who want to use machine learning.

Incorrect options:

Amazon Polly - You can use Amazon Polly to turn text into lifelike speech thereby allowing you to create applications that talk. Polly's Text-to-Speech (TTS) service uses advanced deep learning technologies to synthesize natural sounding human speech.

Amazon Comprehend - Amazon Comprehend is a natural language processing (NLP) service that uses machine learning to find meaning and insights in text. Natural Language Processing (NLP) is a way for computers to analyze, understand, and derive meaning from textual information in a smart and useful way. By utilizing natural language processing (NLP), you can extract important phrases, sentiment,

syntax, key entities such as brand, date, location, person, etc., and the language of the text.

Amazon Connect - Amazon Connect is an omnichannel cloud contact center. You can set up a contact center in a few steps, add agents who are located anywhere, and start engaging with your customers. You can create personalized experiences for your customers using omnichannel communications. Amazon Connect is an open platform that you can integrate with other enterprise applications.

Reference:

<https://aws.amazon.com/sagemaker/>

Domain

Technology

Question 58Correct

Which AWS service can inspect Amazon CloudFront distributions running on any HTTP web server?

Your answer is correct

AWS Web Application Firewall (AWS WAF)

AWS GuardDuty

Amazon Inspector

Elastic Load Balancing (ELB)

Overall explanation

Correct option:

AWS Web Application Firewall (AWS WAF)

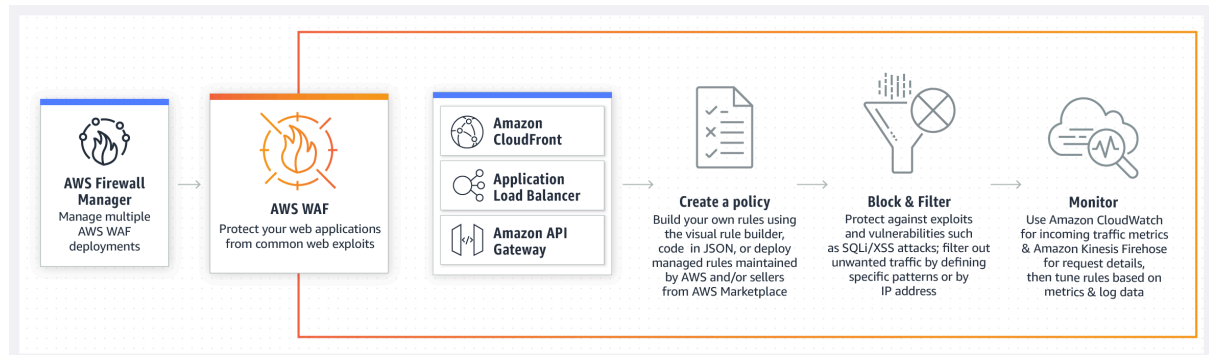
AWS Web Application Firewall (AWS WAF) is a web application firewall that helps protect web applications from attacks by allowing you to configure rules that allow, block, or monitor (count) web requests based on conditions that you define. These conditions include IP addresses, HTTP headers, HTTP body, URI strings, SQL injection, and cross-site scripting (XSS).

AWS WAF is a web application firewall that lets you monitor the HTTP and HTTPS requests that are forwarded to Amazon CloudFront and lets you control access to your content.

When you use the AWS web application firewall (AWS WAF) on Amazon CloudFront, your rules run in all AWS Edge Locations, located around the world close to your

end-users. This means security doesn't come at the expense of performance. Blocked requests are stopped before they reach your web servers.

How AWS WAF works:



via - <https://aws.amazon.com/waf/>

Incorrect options:

AWS GuardDuty - Amazon GuardDuty is a threat detection service that continuously monitors for malicious or unauthorized behavior to help you protect your AWS accounts and workloads. It does not inspect Amazon CloudFront distributions.

Amazon Inspector - Amazon Inspector is an automated security assessment service that helps improve the security and compliance of applications deployed on AWS. Amazon Inspector automatically assesses applications for exposure, vulnerabilities, and deviations from best practices. Amazon Inspector security assessments help you check for unintended network accessibility of your Amazon EC2 instances and for vulnerabilities on those EC2 instances.

Elastic Load Balancing (ELB) - Elastic Load Balancing (ELB) automatically distributes incoming application traffic across multiple targets, such as Amazon EC2 instances, containers, and IP addresses. It does not inspect Amazon CloudFront distributions.

Reference:

<https://aws.amazon.com/waf/>

Domain

Technology

Question 59Correct

Which of the following AWS Identity and Access Management (AWS IAM) Security Tools allows you to review permissions granted to an IAM user?

Multi-Factor Authentication (MFA)

IAM credentials report

Your answer is correct

AWS Identity and Access Management (IAM) access advisor

IAM policy

Overall explanation

Correct option:

AWS Identity and Access Management (IAM) access advisor

IAM Access advisor shows the service permissions granted to a user and when those services were last accessed. You can use this information to revise your policies. To summarize, you can identify unnecessary permissions so that you can revise your IAM policies accordingly.

Incorrect options:

IAM credentials report - You can generate and download a credential report that lists all IAM users in your account and the status of their various credentials, including passwords, access keys, and multi-factor authentication (MFA) devices. It is not used to review permissions granted to an IAM user.

IAM policy - IAM policies define permissions for an action regardless of the method that you use to perform the operation.

Multi-Factor Authentication (MFA) - Multi-Factor Authentication (MFA) is a simple best practice that adds an extra layer of protection on top of your user name and password. With Multi-Factor Authentication (MFA) enabled, when a user signs in to an AWS Management Console, they will be prompted for their user name and password (the first factor—what they know), as well as for an authentication code from their AWS MFA device (the second factor—what they have). Taken together, these multiple factors provide increased security for your AWS account settings and resources. It cannot be used to review permissions granted.

Reference:

<https://aws.amazon.com/about-aws/whats-new/2019/06/now-use-iam-access-advisor-with-aws-organizations-to-set-permission-guardrails-confidently/>

Domain

Security and Compliance

Question 60Correct

Which AWS tool/service will help you define your cloud infrastructure using popular programming languages such as Python and JavaScript?

AWS CloudFormation

Your answer is correct

AWS Cloud Development Kit (AWS CDK)

AWS Elastic Beanstalk

AWS CodeBuild

Overall explanation

Correct option:

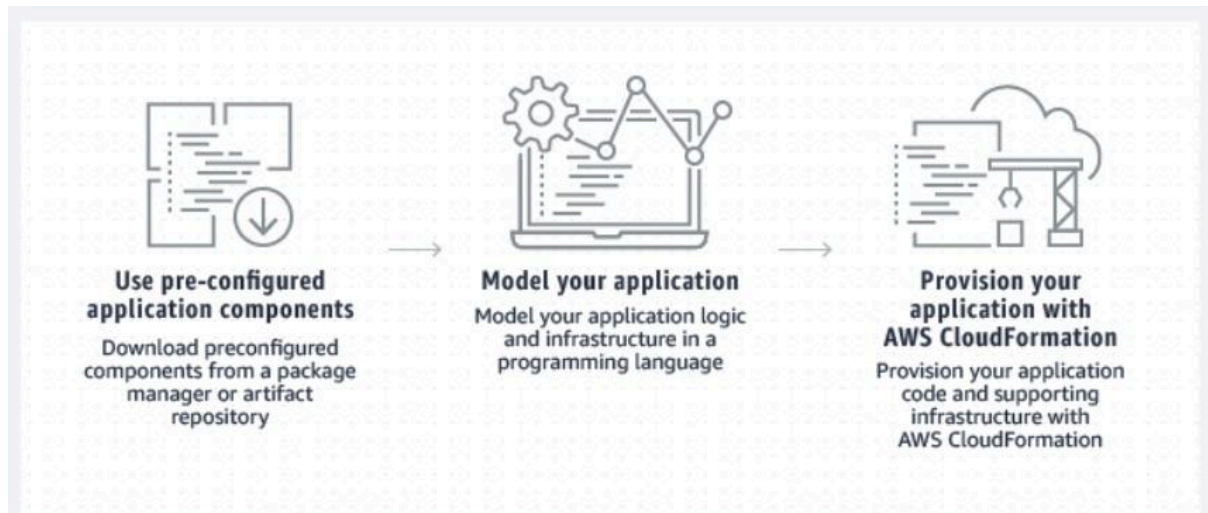
AWS Cloud Development Kit (AWS CDK)

The AWS Cloud Development Kit (AWS CDK) is an open-source software development framework to define your cloud application resources using familiar programming languages.

AWS Cloud Development Kit (AWS CDK) uses the familiarity and expressive power of programming languages for modeling your applications. It provides you with high-level components called constructs that preconfigure cloud resources with proven defaults, so you can build cloud applications without needing to be an expert. AWS CDK provisions your resources in a safe, repeatable manner through AWS CloudFormation. It also enables you to compose and share your own custom constructs that incorporate your organization's requirements, helping you start new projects faster.

In short, you use the AWS CDK framework to author AWS CDK projects which are executed to generate AWS CloudFormation templates.

How Cloud Development Kit (AWS CDK) works:



via - <https://aws.amazon.com/cdk/>

Incorrect options:

AWS Elastic Beanstalk - AWS Elastic Beanstalk is an easy-to-use service for deploying and scaling web applications and services developed with Java, .NET, PHP, Node.js, Python, etc. You can simply upload your code in a programming language of your choice and AWS Elastic Beanstalk automatically handles the deployment, from capacity provisioning, load balancing, and auto scaling to application health monitoring.

AWS CloudFormation - AWS CloudFormation is a service that gives developers and businesses an easy way to create a collection of related AWS and third-party resources, and provision and manage them in an orderly and predictable fashion. AWS CloudFormation is designed to allow resource lifecycles to be managed repeatably, predictably, and safely while allowing for automatic rollbacks, automated state management, and management of resources across accounts and regions. AWS Cloud Development Kit (AWS CDK) helps code the same in higher-level languages and converts them into AWS CloudFormation templates.

AWS CodeBuild - AWS CodeBuild is a fully managed continuous integration service that compiles source code, runs tests, and produces software packages that are ready to deploy. With AWS CodeBuild, you don't need to provision, manage, and scale your own build servers. AWS CodeBuild scales continuously and processes multiple builds concurrently, so your builds are not left waiting in a queue.

Reference:

<https://aws.amazon.com/cdk/>

Domain

Technology

Question 61 Correct

A production company would like to establish an AWS managed virtual private network (VPN) service between its on-premises network and AWS. Which item needs to be set up on the company's side?

A virtual private gateway (VGW)

A security group

Your answer is correct

A customer gateway

A VPC endpoint interface

Overall explanation

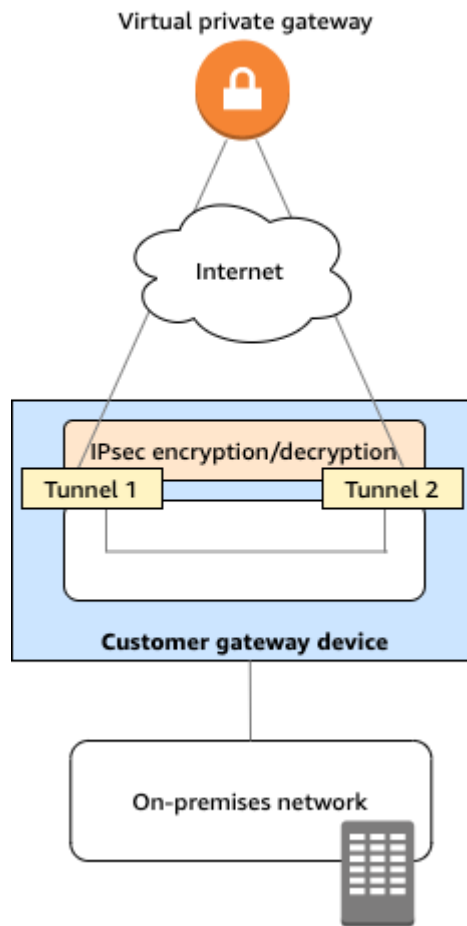
Correct option:

A customer gateway

A customer gateway device is a physical or software appliance on your side of a Site-to-Site VPN connection. You or your network administrator must configure the device to work with the Site-to-Site VPN connection.

You can enable access to your remote network from your VPC by creating an AWS Site-to-Site VPN (Site-to-Site VPN) connection, and configuring routing to pass traffic through the connection.

More on customer gateway device:



<https://docs.aws.amazon.com/vpn/latest/s2svpn/your-cgw.html>

Incorrect options:

A security group - A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. It is not a component of a connection between on-premises network and AWS.

A VPC endpoint interface - An interface VPC endpoint (interface endpoint) enables you to connect to services powered by AWS PrivateLink. It is not a component of a connection between on-premises network and AWS.

A virtual private gateway (VGW) - A virtual private gateway (VGW) device is a physical or software appliance on AWS side of a Site-to-Site VPN connection.

References:

<https://docs.aws.amazon.com/vpn/latest/s2svpn/your-cgw.html>

https://docs.aws.amazon.com/vpn/latest/s2svpn/VPC_VPN.html

Domain

Technology

Question 62Correct

According to the AWS Shared Responsibility Model, which of the following is the responsibility of the customer?

Edge locations security

Protecting hardware infrastructure

Managing Amazon DynamoDB

Your answer is correct

Firewall & networking configuration of Amazon Elastic Compute Cloud (Amazon EC2)

Overall explanation

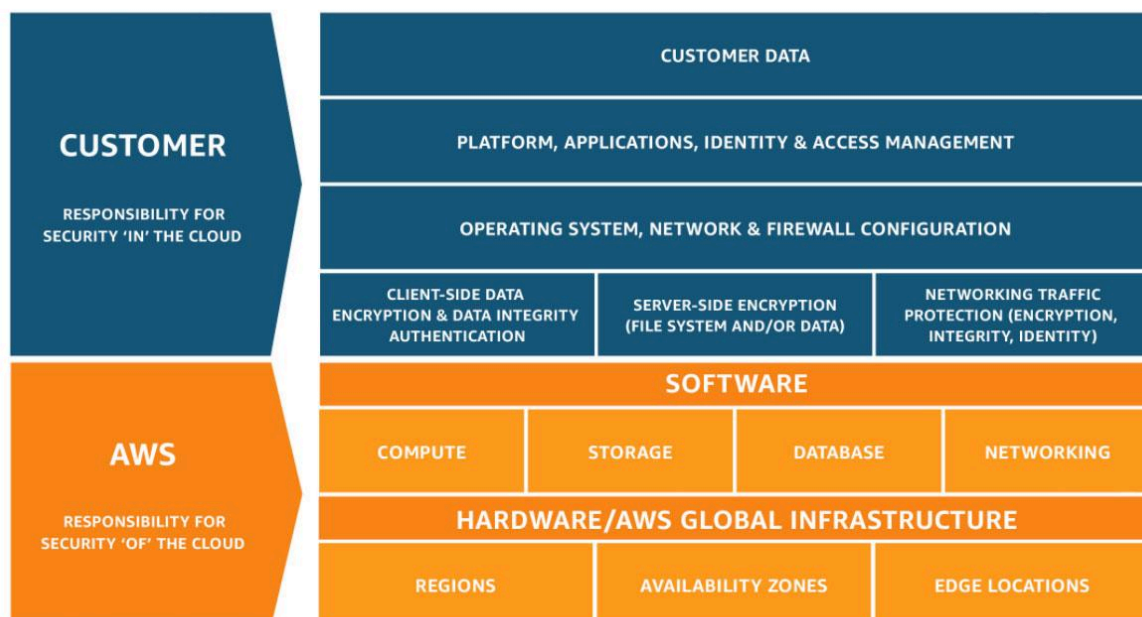
Correct option:

Firewall & networking configuration of Amazon Elastic Compute Cloud (Amazon EC2)

The customers are responsible for "Security IN the cloud". It includes the configuration of the operating system, network & firewall of applications.

Exam Alert:

Please review the AWS Shared Responsibility Model in detail as you can expect multiple questions on the shared responsibility model in the exam:



via - <https://aws.amazon.com/compliance/shared-responsibility-model/>

Incorrect options:

Managing Amazon DynamoDB - Amazon DynamoDB is a fully managed service. AWS operates the infrastructure layer, the operating system, and platforms, and customers access the endpoints to store and retrieve data.

Protecting hardware infrastructure

Edge locations security

AWS is responsible for "Security OF the cloud". It includes the infrastructure, which is composed of the hardware, software, networking, and facilities that run AWS Cloud services.

Reference:

<https://aws.amazon.com/compliance/shared-responsibility-model/>

Domain

Security and Compliance

Question 63Correct

The development team at a company manages 300 microservices and it is now trying to automate the code reviews to improve the code quality. Which tool/service is the right fit for this requirement?

AWS CodeBuild

Your answer is correct

Amazon CodeGuru

AWS X-Ray

AWS Trusted Advisor

Overall explanation

Correct option:

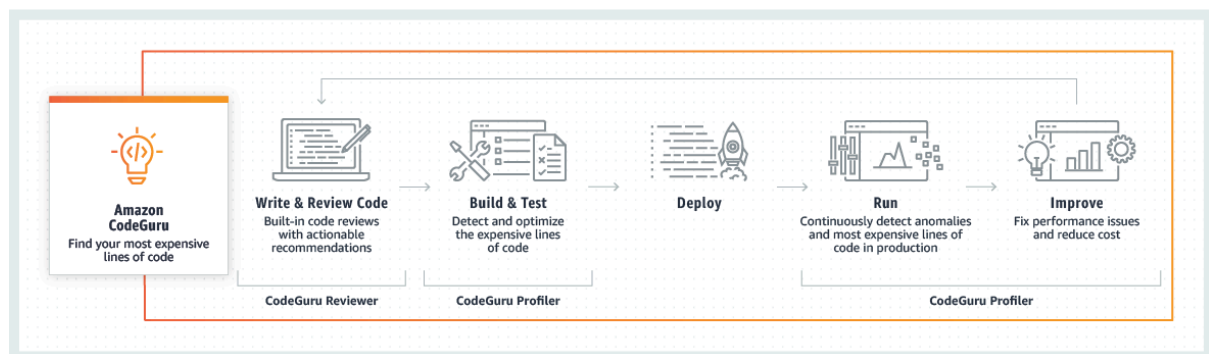
Amazon CodeGuru

Amazon CodeGuru is a developer tool that provides intelligent recommendations to improve code quality and identify an application's most expensive lines of code. Integrate Amazon CodeGuru into your existing software development workflow to automate code reviews during application development, continuously monitor application performance in production, provide recommendations and visual clues for improving code quality and application performance, and reduce overall cost.

Amazon CodeGuru Reviewer uses machine learning and automated reasoning to identify critical issues, security vulnerabilities, and hard-to-find bugs during application development and provides recommendations to improve code quality.

Amazon CodeGuru Profiler pinpoints an application's most expensive lines of code by helping developers understand the runtime behavior of their applications, identify and remove code inefficiencies, improve performance, and significantly decrease compute costs.

How Amazon CodeGuru works:



via - <https://aws.amazon.com/codeguru/>

Incorrect options:

AWS X-Ray - AWS X-Ray helps developers analyze and debug production, and distributed applications, such as those built using a microservices architecture. With AWS X-Ray, you can understand how your application and its underlying services are performing to identify and troubleshoot the root cause of performance issues and errors. AWS X-Ray provides an end-to-end view of requests as they travel through your application, and shows a map of your application's underlying components.

AWS CodeBuild - AWS CodeBuild is a fully managed continuous integration service that compiles source code, runs tests, and produces software packages that are ready to deploy. With AWS CodeBuild, you don't need to provision, manage, and scale your own build servers. AWS CodeBuild scales continuously and processes multiple builds concurrently, so your builds are not left waiting in a queue.

AWS Trusted Advisor - AWS Trusted Advisors provides recommendations that help you follow AWS best practices. AWS Trusted Advisor evaluates your account by

using checks. These checks identify ways to optimize your AWS infrastructure, improve security and performance, reduce costs, and monitor service quotas.

Reference:

<https://aws.amazon.com/codeguru/>

Domain

Technology

Question 64Correct

Which service/tool will you use to create and provide trusted users with temporary security credentials that can control access to your AWS resources?

Amazon Cognito

AWS Web Application Firewall (AWS WAF)

AWS IAM Identity Center

Your answer is correct

AWS Security Token Service (AWS STS)

Overall explanation

Correct option:

AWS Security Token Service (AWS STS)

AWS Security Token Service (AWS STS) is a web service that enables you to request temporary, limited-privilege credentials for AWS Identity and Access Management (AWS IAM) users or for users that you authenticate (federated users).

You can use the AWS Security Token Service (AWS STS) to create and provide trusted users with temporary security credentials that can control access to your AWS resources. Temporary security credentials work almost identically to the long-term access key credentials that your IAM users can use, with the following differences:

1. Temporary security credentials are short-term, as the name implies. They can be configured to last for anywhere from a few minutes to several hours. After the credentials expire, AWS no longer recognizes them or allows any kind of access from API requests made with them.
2. Temporary security credentials are not stored with the user but are generated dynamically and provided to the user when requested. When (or even before) the temporary security credentials expire, the user can request new credentials, as long as the user requesting them still has permission to do so.

Temporary security credentials are generated by AWS Security Token Service (AWS STS). By default, AWS STS is a global service with a single endpoint at <https://sts.amazonaws.com>. However, you can also choose to make AWS STS API calls to endpoints in any other supported Region.

Incorrect options:

Amazon Cognito - Amazon Cognito is a higher level of abstraction than AWS Security Token Service (AWS STS). Amazon Cognito supports the same identity providers as AWS STS, and also supports unauthenticated (guest) access, and lets you migrate user data when a user signs in. Amazon Cognito also provides API operations for synchronizing user data so that it is preserved as users move between devices. Amazon Cognito helps create the user database, which is not possible with STS.

AWS IAM Identity Center - AWS IAM Identity Center is the successor to AWS Single Sign-On (AWS SSO). It is built on top of AWS Identity and Access Management (AWS IAM) to simplify access management to multiple AWS accounts, AWS applications, and other SAML-enabled cloud applications. In AWS IAM Identity Center, you create or connect, your workforce users for use across AWS. You can choose to manage access just to your AWS accounts, just to your cloud applications, or to both.

AWS Web Application Firewall (AWS WAF) - AWS Web Application Firewall (AWS WAF) is a web application firewall that helps protect your web applications or APIs against common web exploits and bots that may affect availability, compromise security, or consume excessive resources. AWS WAF gives you control over how traffic reaches your applications by enabling you to create security rules that control bot traffic and block common attack patterns, such as SQL injection or cross-site scripting.

Reference:

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_temp.html

Domain

Security and Compliance

Question 65Correct

A company would like to move its infrastructure to AWS Cloud. Which of the following should be included in the Total Cost of Ownership (TCO) estimate? (Select TWO)

Your selection is correct

Power/Cooling

Number of end-users

Electronic equipment at office

Application advertising

Your selection is correct

Server administration

Overall explanation

Correct options:

Server administration

Power/Cooling

AWS Pricing Calculator lets you explore AWS services and create an estimate for the cost of your use cases on AWS. You can model your solutions before building them, explore the price points and calculations behind your estimate, and find the available instance types and contract terms that meet your needs. This enables you to make informed decisions about using AWS. You can plan your AWS costs and usage or price out by setting up a new set of instances and services. AWS Pricing Calculator can be accessed at <https://calculator.aws/#/>.

AWS Pricing Calculator compares the cost of your applications in an on-premises or traditional hosting environment to AWS: server, storage, network, and IT labor.

Therefore, you need to include every element relevant to these points of comparison.

Server administration is included in the IT labor costs.

Power/Cooling are included in the server, storage, and network cost.

Incorrect options:

Application advertising - The application advertising is not relevant for a Total Cost of Ownership (TCO) estimate.

Number of end-users - The number of end-users is not relevant for a Total Cost of Ownership (TCO) estimate.

Electronic equipment at office - The electronic equipment at the office is not relevant for a Total Cost of Ownership (TCO) estimate.

References:

<https://calculator.aws/#/>

<https://aws.amazon.com/blogs/aws/new-cloud-tco-comparison-calculator-for-web-applications/>

Domain

Billing and Pricing